Computational Sociology

LLMs in Social Science Research

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November 6, 2026

Plan

1. Course updates
2. Generative AI and social science
3. Measurement
4. Simulation
5. Human–AI experiments
6. Research practice

Course updates

Project progress report

- Progress report due today
- Submit PDF via email

Course updates

Homework 3

- Homework 3 due next Friday, November 13

Course updates

Presentations

- Student presentations on Friday, November 20
- 12-minute presentation, plus Q&A

Generative AI and social science

Generative AI and social science

Opportunities and risks

- Generative AI has the potential to improve survey research, online experiments, automated content analyses, agent-based models, and other techniques used to study human behavior
- Bias, ethics, replication, environmental impact, and the proliferation of low-quality research

Bail 2024

Generative AI and social science

Measurement, prompting, and simulation

- *Measurement*: Rigorous measurement and validation of AI-generated outputs
- *Prompting*: Optimizing model performance and reproducibility
- *Simulation*: Novel uses of AI to simulate attitudes and behaviors

Davidson and Karell 2025

Measurement

Text analysis with LLMs

Methodological bricolage

- New opportunities for "methodological bricolage", as the same model can be used in multiple capacities
  - e.g. LLM as classifier, topic model, and embedding
- Enables rapid prototyping, experimentation, and bespoke solutions, making computational text analysis more flexible and accessible

Bonikowski and Nelson 2022; Davidson 2024

Text analysis with LLMs

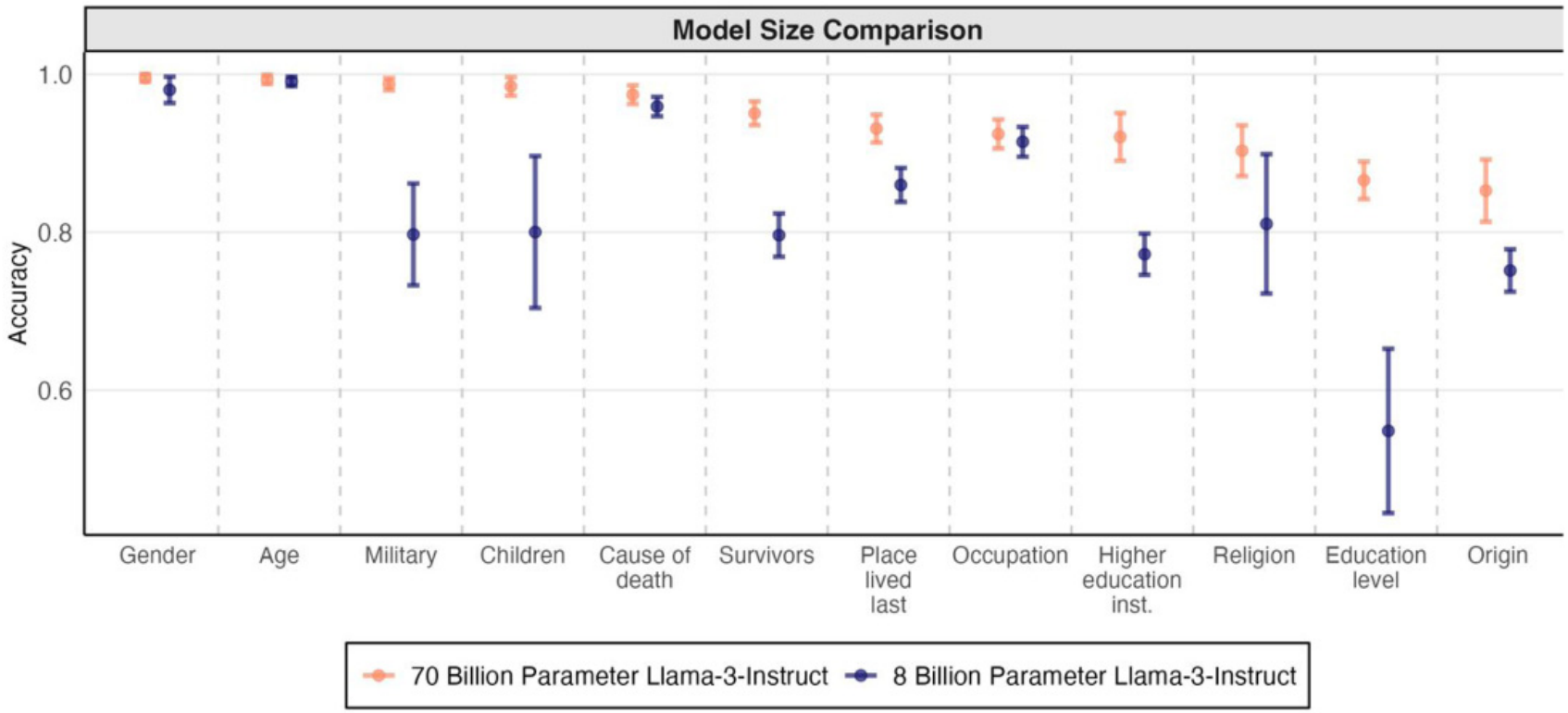
From codebooks to promptbooks

- LLMs can be prompted to extract structured information from text
  - e.g. Age, education, military service, and number of children from *New York Times* obituaries
- A *promptbook* is "the complete set of prompts used in a project", analogous to a codebook
  - Prompts are revised on a development set without changing the concept being measured

Stuhler, Oscar, Cat Dang Ton, and Étienne Ollion. 2025. "From Codebooks to Promptbooks: Extracting Information from Text with Generative Large Language Models." *Sociological Methods & Research* 54(3): 794–848.

Text analysis with LLMs

Information extraction



Explicit information near perfect, but lower accuracy when interpretation needed

Stuhler et al. 2025

Text analysis with LLMs

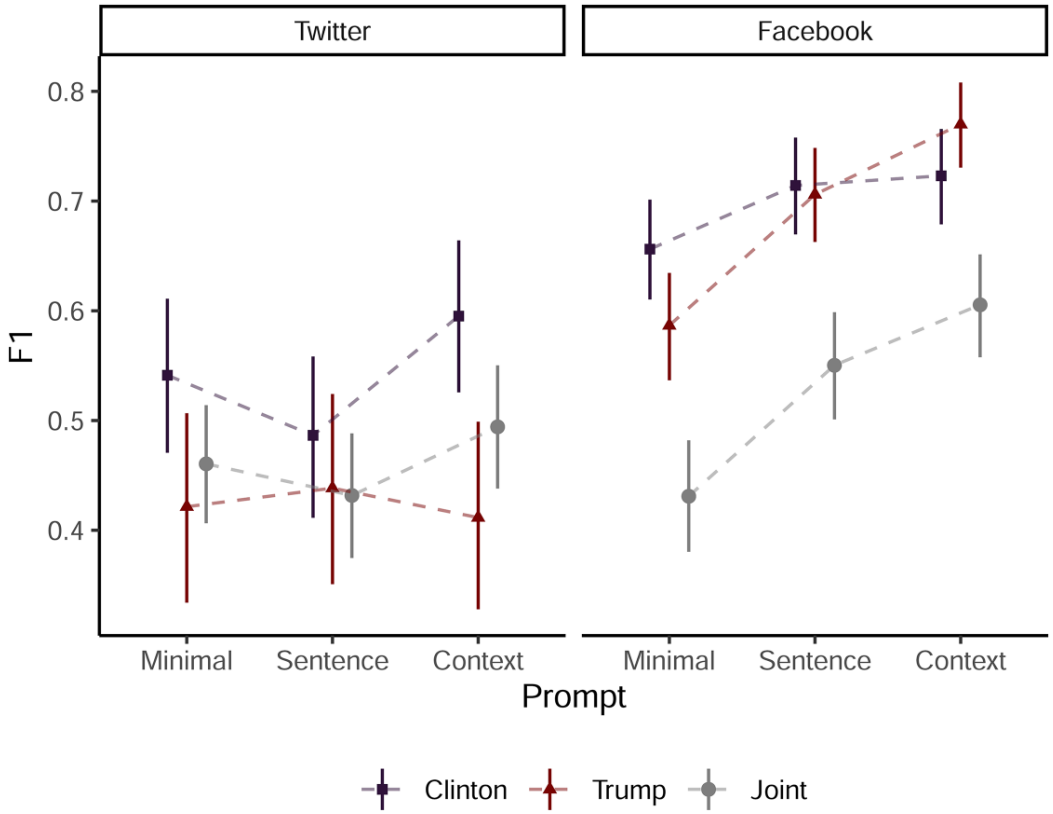
Text classification with LLMs

- How does performance vary across models and learning regimes?
- Ten models, from BERT to GPT-4o, and four learning regimes
  - *Zero-shot* and *few-shot* learning
  - *Fine-tuning*
  - *Instruction-tuning*: Fine-tuning a generative model on a prompt and input-output pairs

Chae and Davidson 2026

Text analysis with LLMs

Sensitivity to prompts

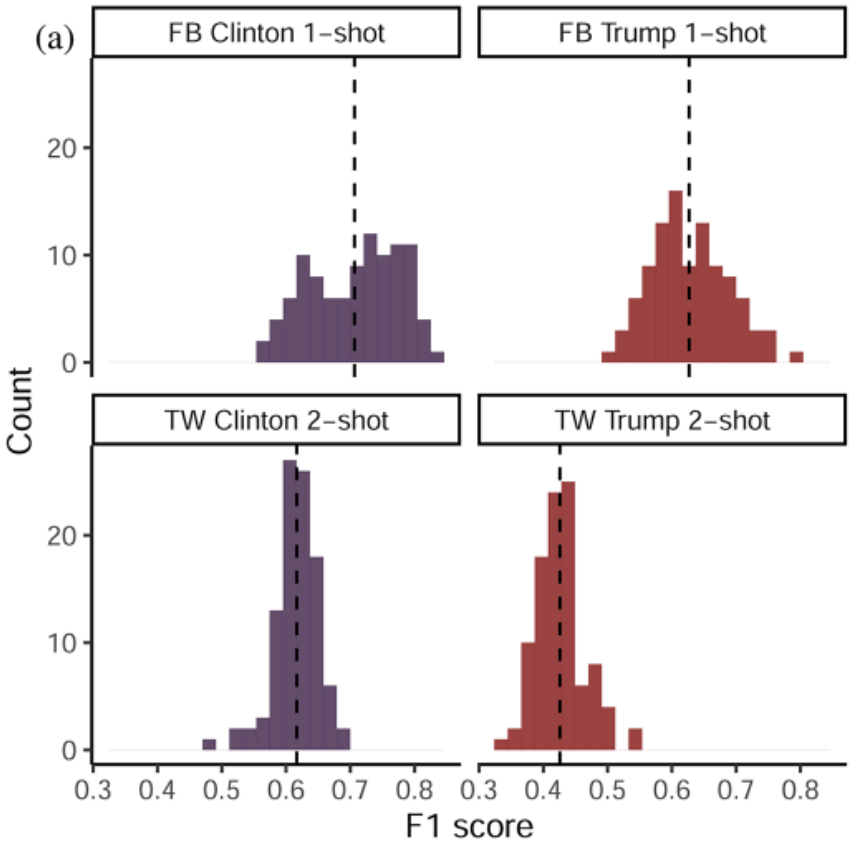


Zero-shot accuracy on GPT-3 across prompts

Chae and Davidson 2026

Text analysis with LLMs

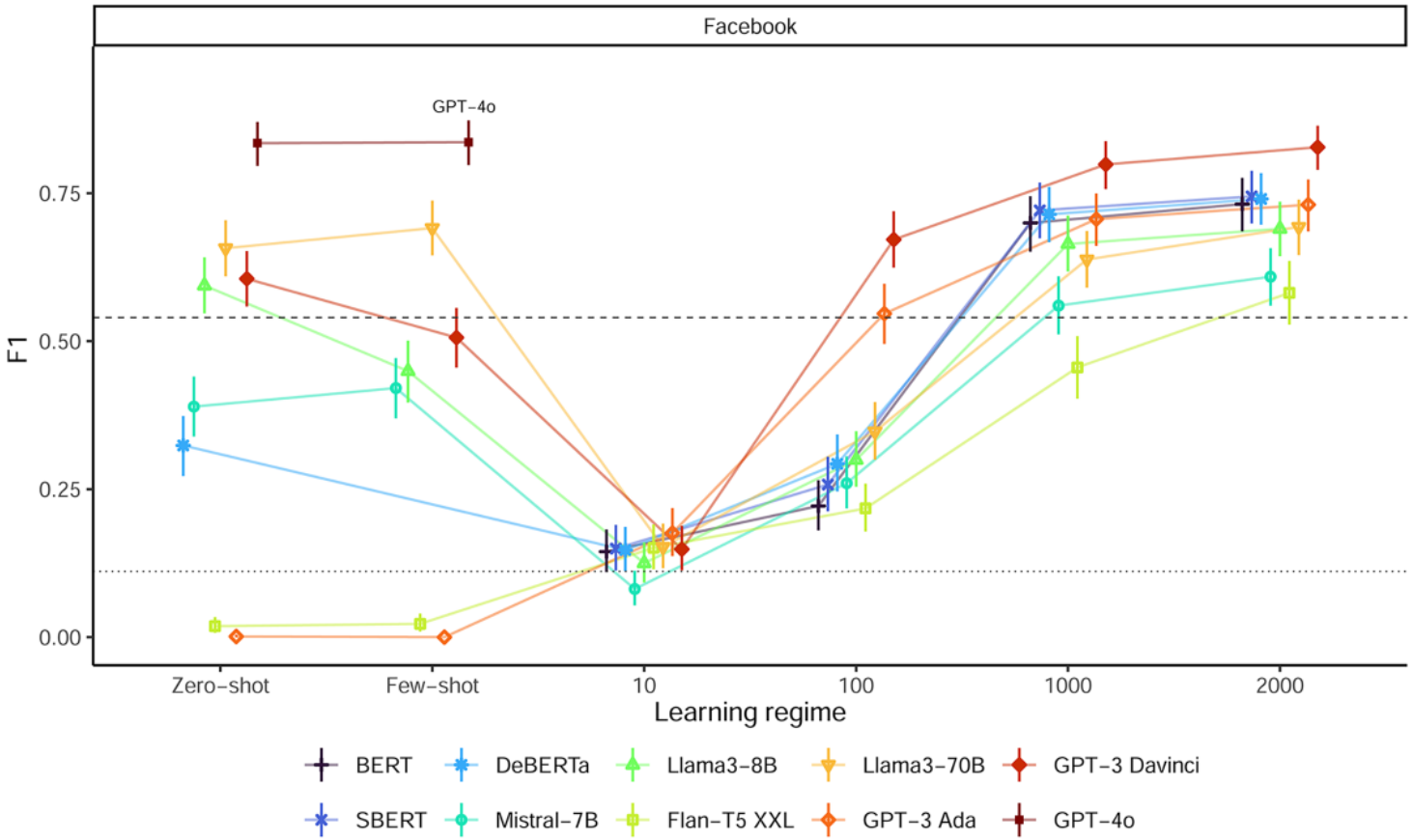
Sensitivity to examples



Chae and Davidson 2026

Text analysis with LLMs

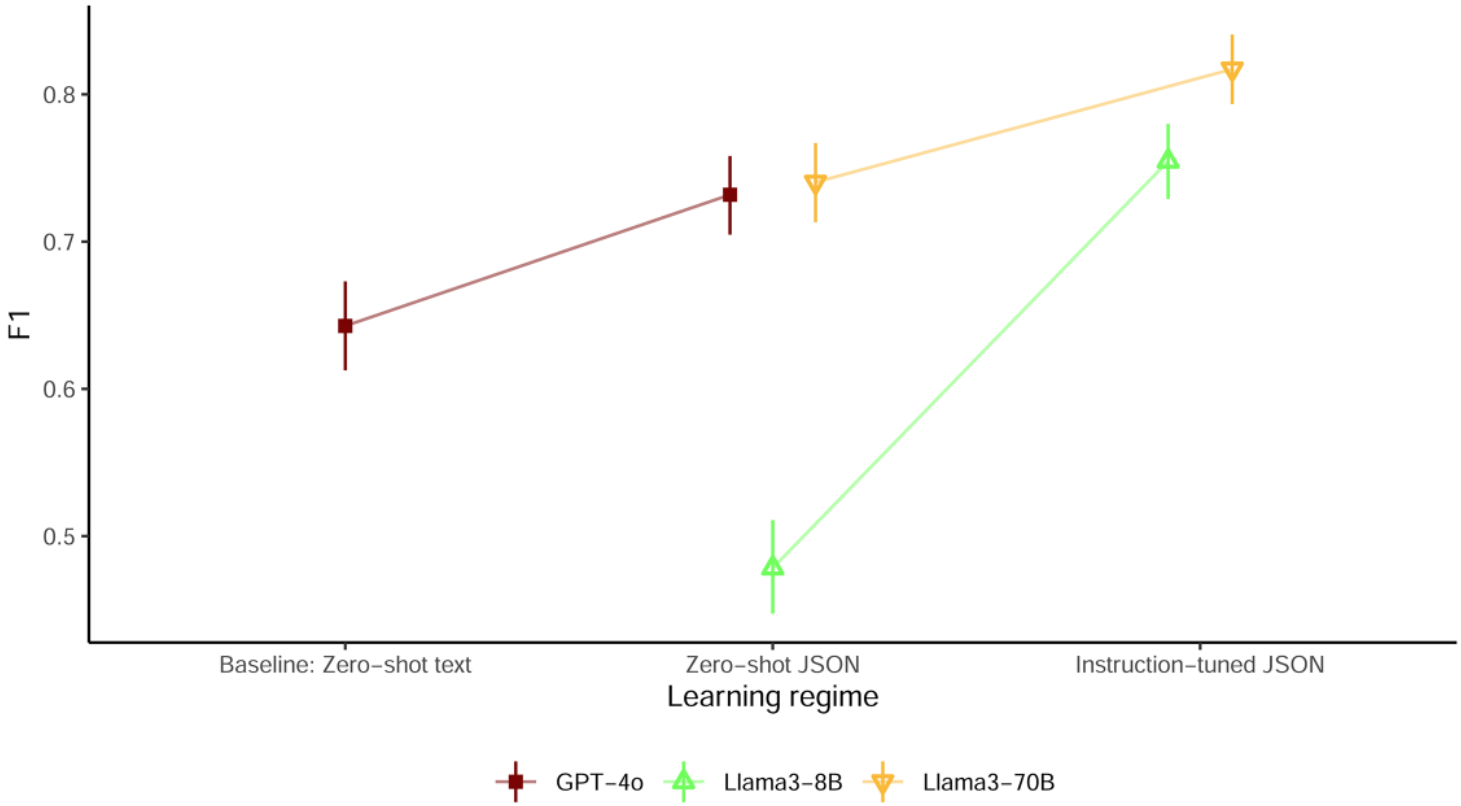
Models and learning regimes



Chae and Davidson 2026

Text analysis with LLMs

Instruction-tuning



Chae and Davidson 2026

Measurement error

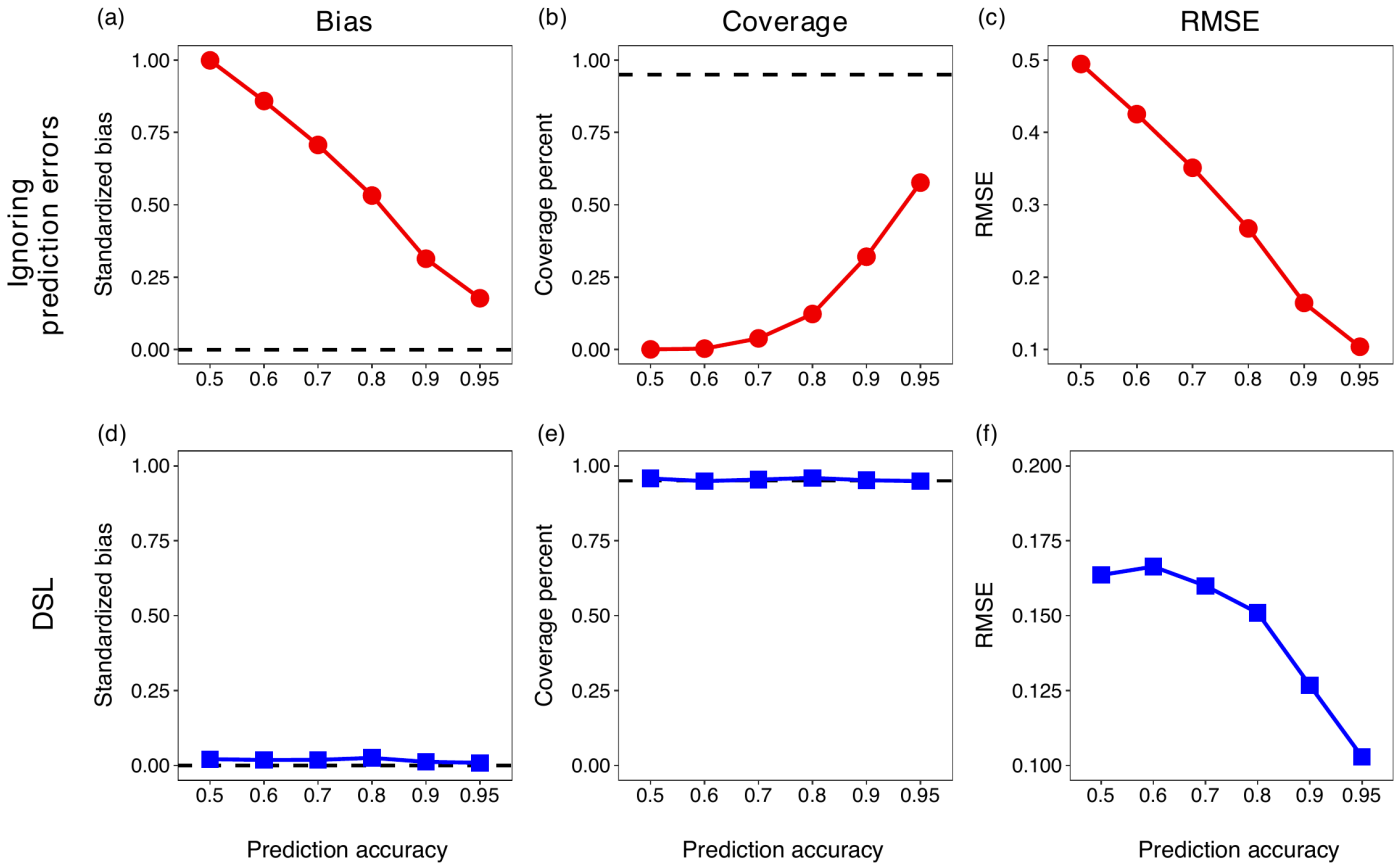
Correcting measurement errors

- Using predictions can result in biased estimates and incorrect confidence intervals
- Methods such as *design-based supervised learning (DSL)* adjust for classifier error

Egami, Naoki, Musashi Hinck, Brandon M. Stewart, and Hanying Wei. 2024. "Using Large Language Model Annotations for the Social Sciences: A General Framework of Using Predicted Variables in Downstream Analyses." Working paper.

Measurement error

Correcting measurement errors



Egami et al. 2024

Qualitative analysis with LLMs

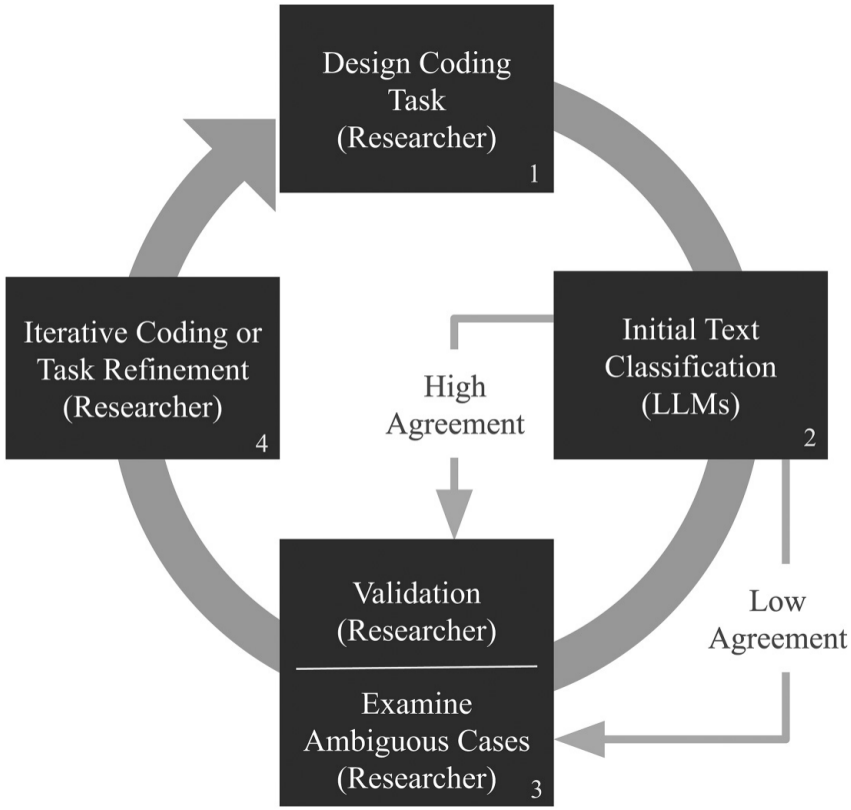
Computational methods and qualitative research

- Generative AI has several advantages over conventional computational methods:
  - Input texts need not be comparable or standardized
  - Queries can be tailored to the task
  - Applications for many tasks, including transcription, translation, exploratory analysis, and coding

Davidson 2024

Qualitative analysis with LLMs

Qualitative coding with LLMs



Than, Nga, Leanne Fan, Tina Law, Laura K. Nelson, and Leslie McCall. 2025. "Updating 'The Future of Coding': Qualitative Coding with Generative Large Language Models." *Sociological Methods & Research* 54(3): 849–88.

Qualitative analysis with LLMs

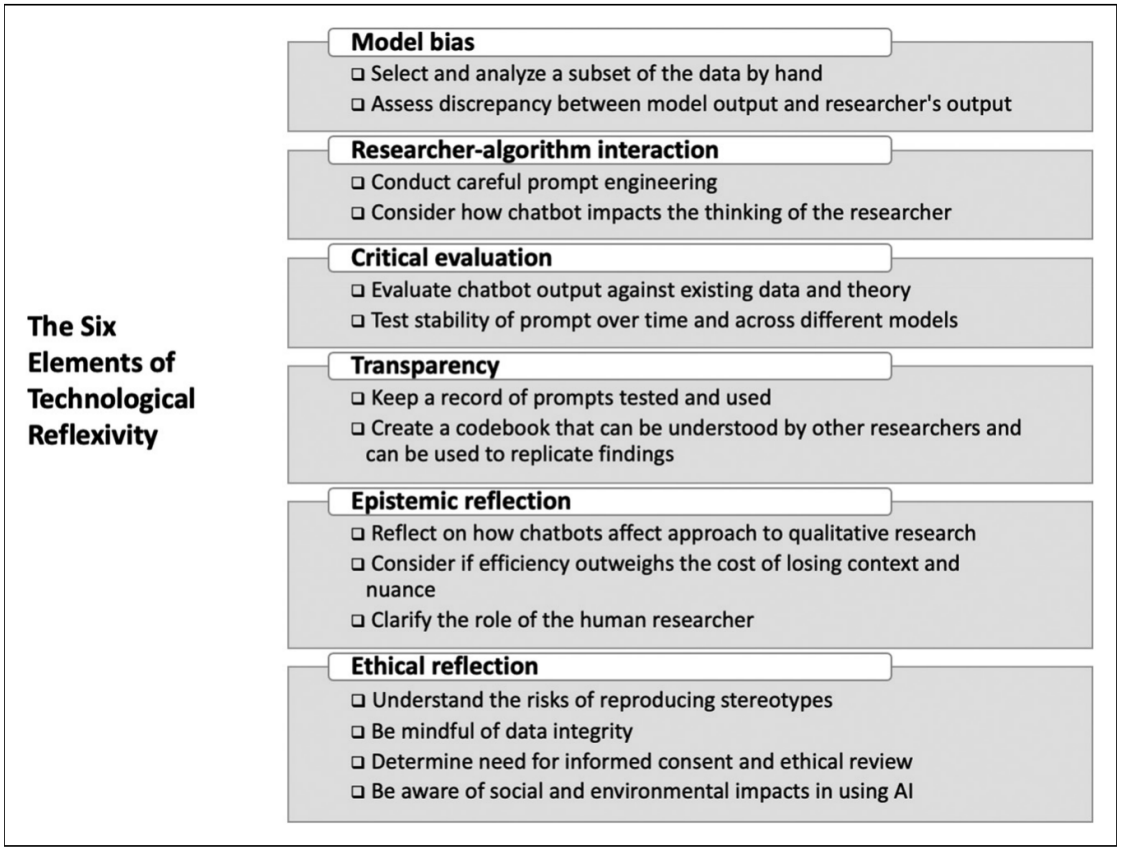
Technological reflexivity

- LLM chatbots are "fundamentally qualitative", as they are built from text and are sensitive to social meaning
- Instead of statistical validity and generalizability, research with LLM chatbots requires *technological reflexivity*
- LLM chatbots cannot replace human researchers because results must still be interpreted qualitatively

Ibrahim and Voyer 2026

Qualitative analysis with LLMs

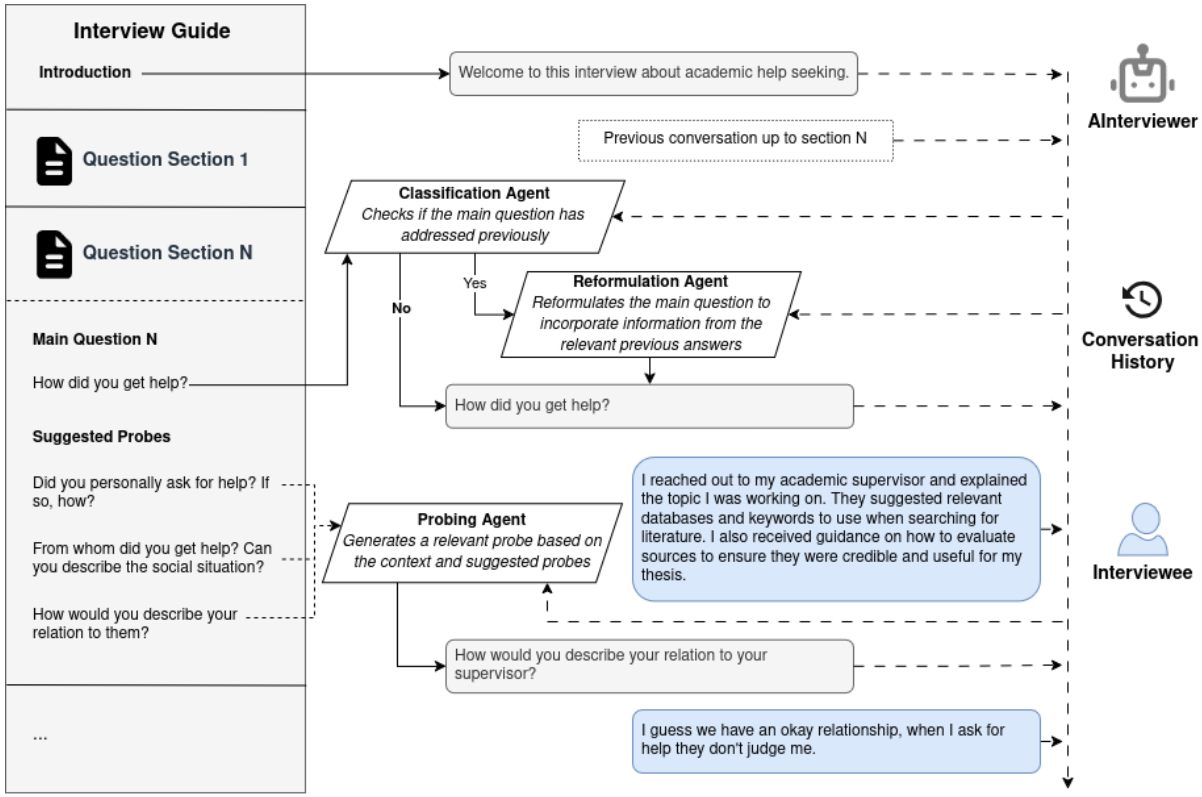
Technological reflexivity



Ibrahim and Voyer 2026

Qualitative analysis with LLMs

AI-led interviews



Gårdhus, Tobias Priesholm, Nikolas Vitsakis, Fie Lejre Frederiksen, Anna Rogers, and Hjalmar Bang Carlsen. 2026. "AInterviewer: A Platform for Designing and Conducting AI-led Qualitative Interviews." In *Proceedings of the ACL*, 119–27.

Qualitative analysis with LLMs

Rejecting generative AI

- 419 qualitative researchers signed an open letter rejecting generative AI for "reflexive" qualitative approaches, such as thematic analysis
  - Generative AI is "simulated intelligence" and "incapable of meaning-making"
  - Qualitative research "should remain a distinctly human practice"
  - Generative AI harms the environment and exploits workers

Jowsey et al. 2025

Qualitative analysis with LLMs

Response: Thematic analysis and theory

- Thematic analysis is vulnerable to AI because it sidelines theory in favor of identifying themes, which LLMs "excel at"
- Theory-rich approaches such as abductive analysis are more resilient
  - A finding only becomes surprising in light of existing theories
  - Whether a contribution is "interesting" is judged by the scholarly community

Timmermans and Tavory 2026

Simulation

Bias and representation

Machine habitus

- Machine learning techniques and LLMs are not neutral but "see" the world in certain ways, analogous to how Bourdieu's "habitus" shapes experience
- LLMs can reflect cognitive schemas present in public culture
- How can these biases be leveraged for sociological research?

Airoldi, Massimo. 2021. *Machine Habitus: Toward a Sociology of Algorithms*. Cambridge: Polity; Arseniev-Koehler and Foster 2022

Bias and representation

Whose viewpoints are represented?

- LLMs can "overrepresent hegemonic viewpoints", reproducing biases and stereotypes from dominant social groups
- After alignment efforts, ChatGPT responds to survey questions similarly to educated liberals
- Models can be prompted to behave like specific social groups

Bender et al. 2021; Martin, John Levi. 2023. "The Ethico-Political Universe of ChatGPT." *Journal of Social Computing* 4(1): 1–11; Argyle et al. 2023

Bias and representation

Forms of bias

- Bias has become a catch-all term describing different things:
  - Statistical deviation from human ground truth
  - Political leanings embedded in model outputs
  - Stereotyping across social categories
  - Discrimination in consequential decisions
- Bias can enter through pre-training data, post-training, and users' prompts
- It is unclear what a neutral, unbiased model would entail

Nelson, Laura K., Daniel Karell, and Thomas Davidson. 2026. "Generative AI and Society." Working paper.

Silicon sampling

Algorithmic fidelity

- *Silicon samples* are created by conditioning a language model on the socio-demographic backstories of real survey respondents
- *Algorithmic fidelity*: "the degree to which the complex patterns of relationships between ideas, attitudes, and socio-cultural contexts within a model accurately mirror those within a range of human sub-populations"

Argyle et al. 2023

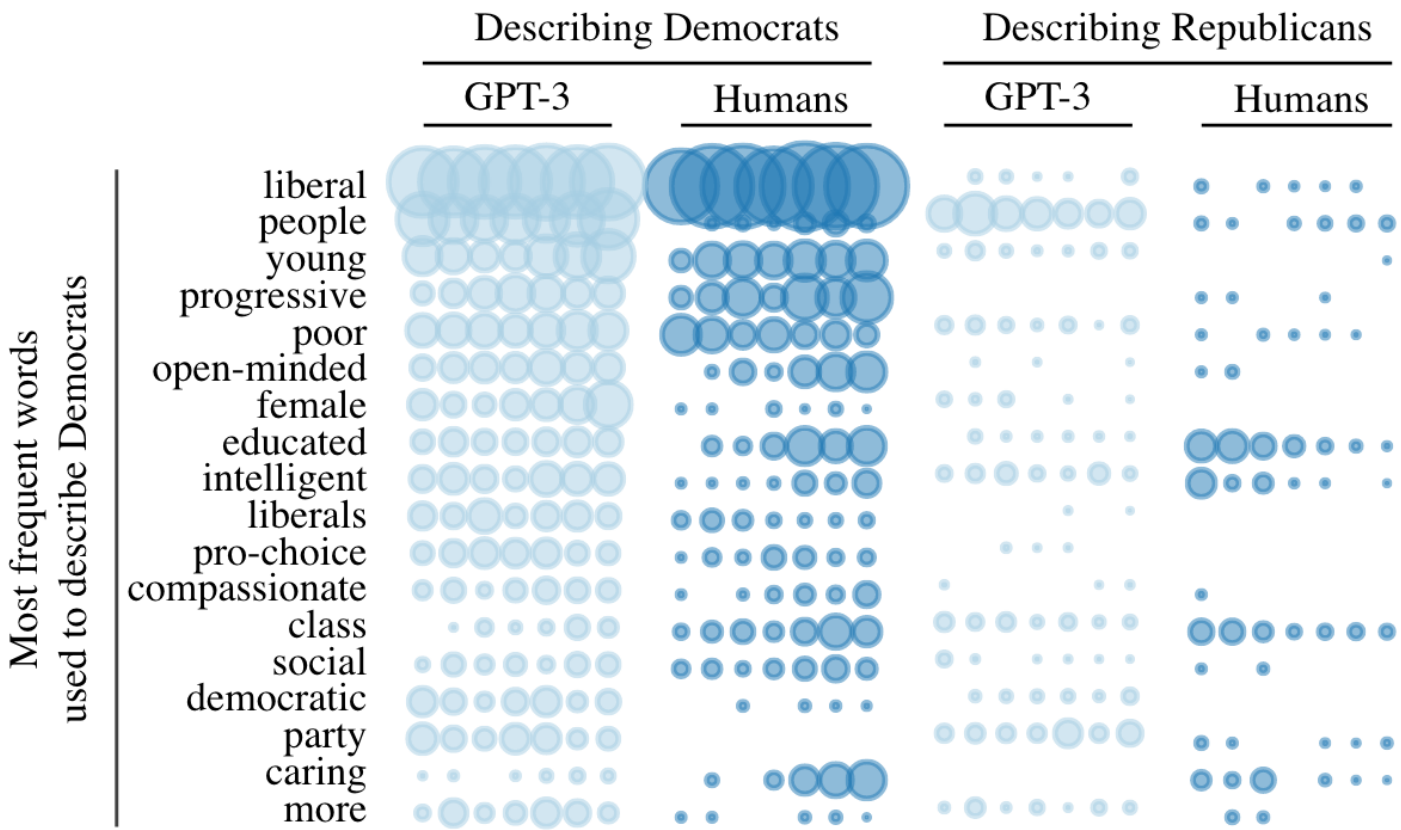
Silicon sampling

Describing partisans

	Describing Democrats	Describing Republicans
Strong Republicans	Ideologically, I describe myself as conservative. Politically, I am a strong Republican. Racially, I am white. I am male. Financially, I am upper-class. In terms of my age, I am young. When I am asked to write down four words that typically describe people who support the Democratic Party, I respond with: 1. **Liberal 2. Socialist 3. Communist 4. Atheist.**	Ideologically, I describe myself as conservative. Politically, I am a strong Republican. Racially, I am white. I am male. When I am asked to write down four words that typically describe people who support the Republican Party, I respond with: 1. **Conservative 2. Male 3. White (or Caucasian) 4. Christian.**
Strong Democrats	Ideologically, I describe myself as liberal. Politically, I am a strong Democrat. Racially, I am white. I am female. Financially, I am poor. In terms of my age, I am old. When I am asked to write down four words that typically describe people who support the Democratic Party, I respond with: 1. **Liberal. 2. Young. 3. Female. 4. Poor.**	Ideologically, I describe myself as extremely liberal. Politically, I am a strong Democrat. Racially, I am hispanic. I am male. Financially, I am upper-class. In terms of my age, I am middle-aged. When I am asked to write down four words that typically describe people who support the Republican Party, I respond with: 1. **Ignorant 2. Racist 3. Misogynist 4. Homophobic.**

Silicon sampling

Describing partisans



Silicon sampling

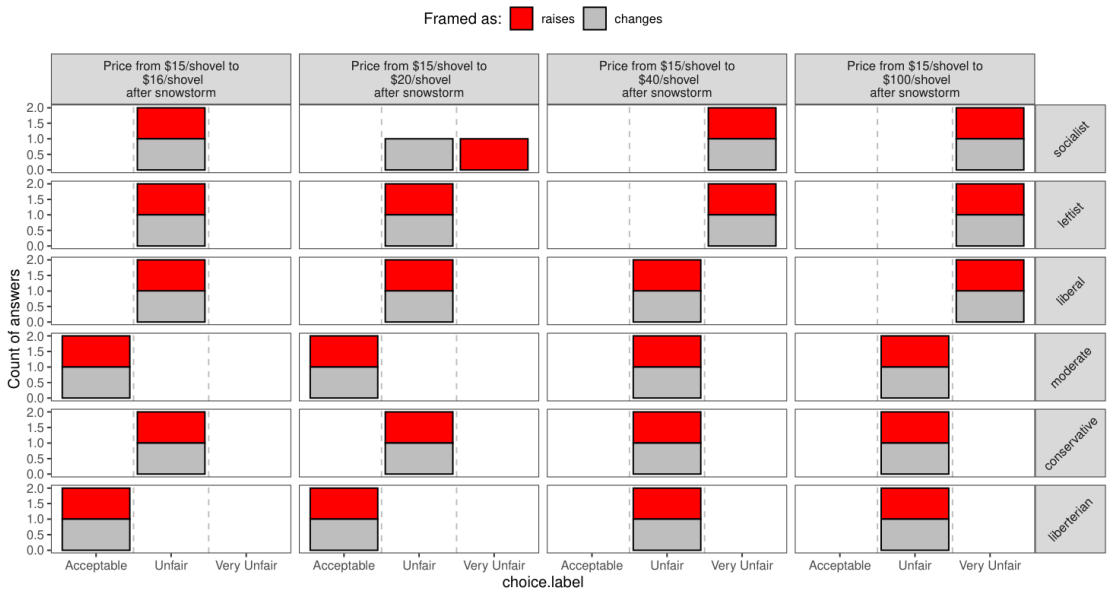
Describing partisans



Silicon sampling

Simulated economic agents

Figure 2: Kahneman et al. (1986) price gouging snow shovel question, with endowed political views

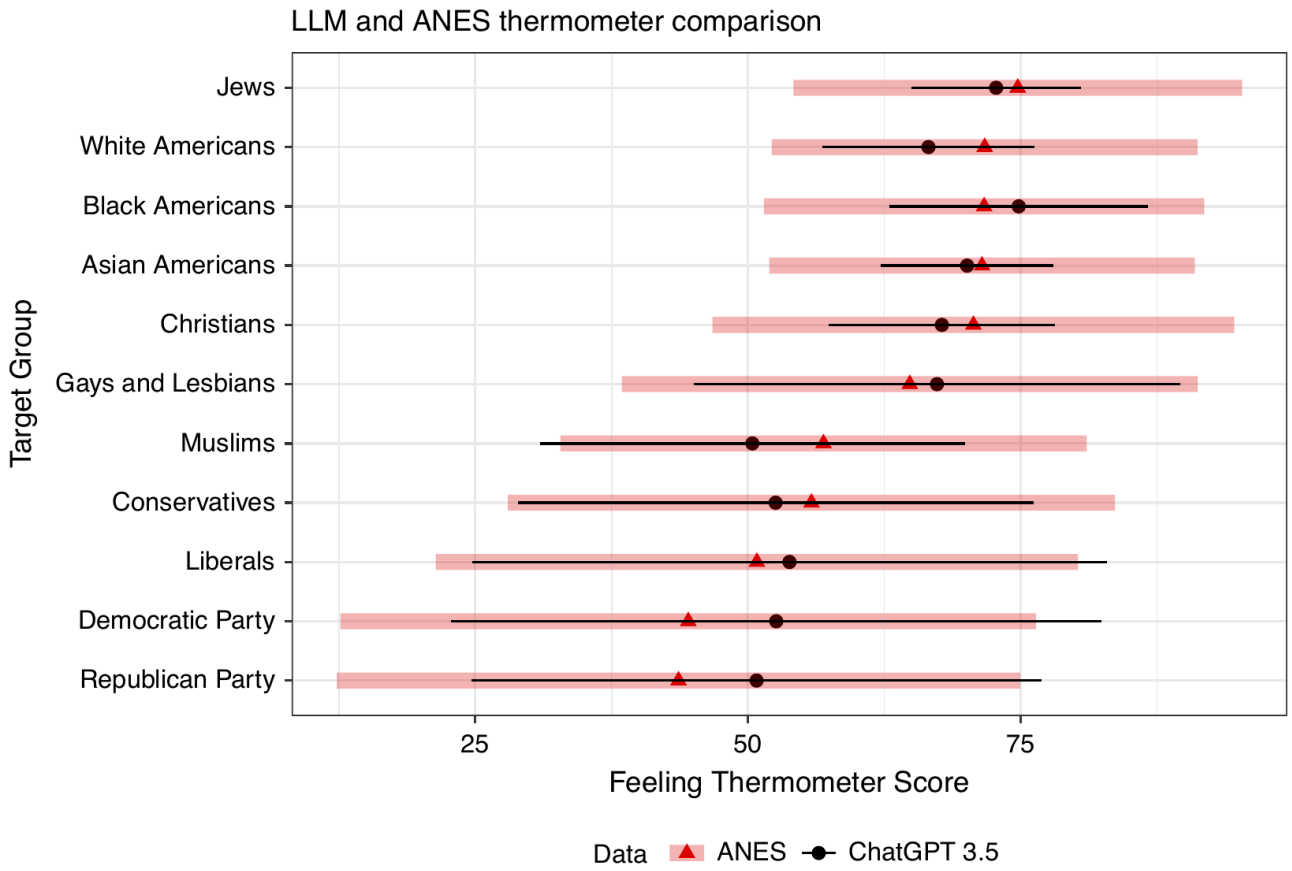


Notes: This shows the faction of AI subjects choosing each more opinion, by scenario.

Horton, John J., Apostolos Filippas, and Benjamin S. Manning. 2023. "Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?" NBER Working Paper 31122.

Silicon sampling

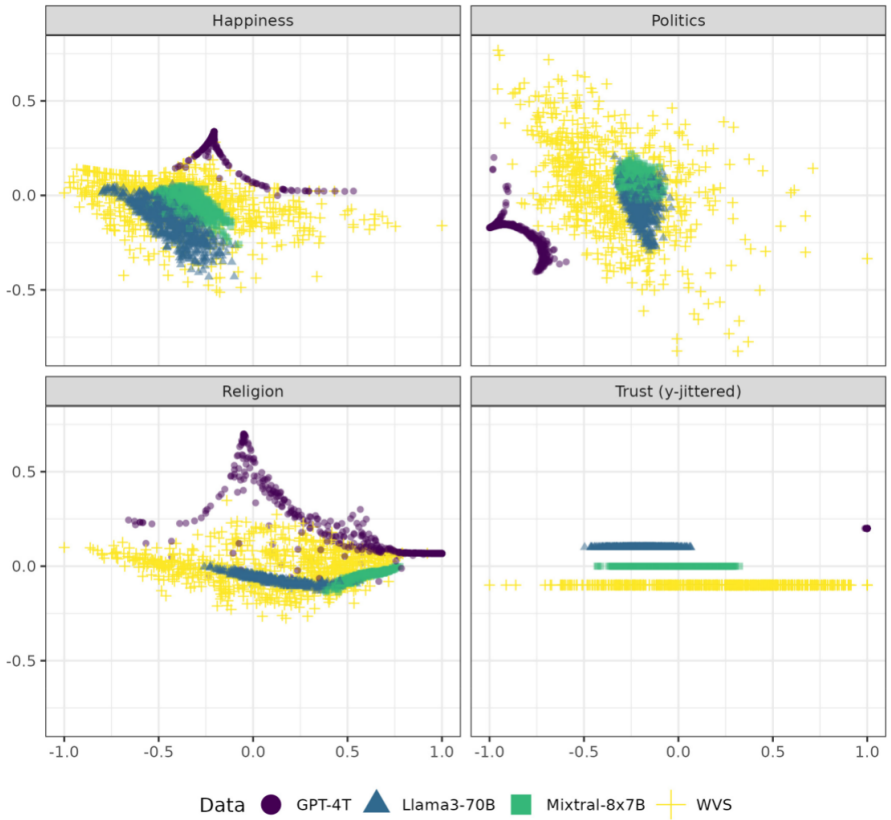
Low variance



Bisbee et al. 2024

Silicon sampling

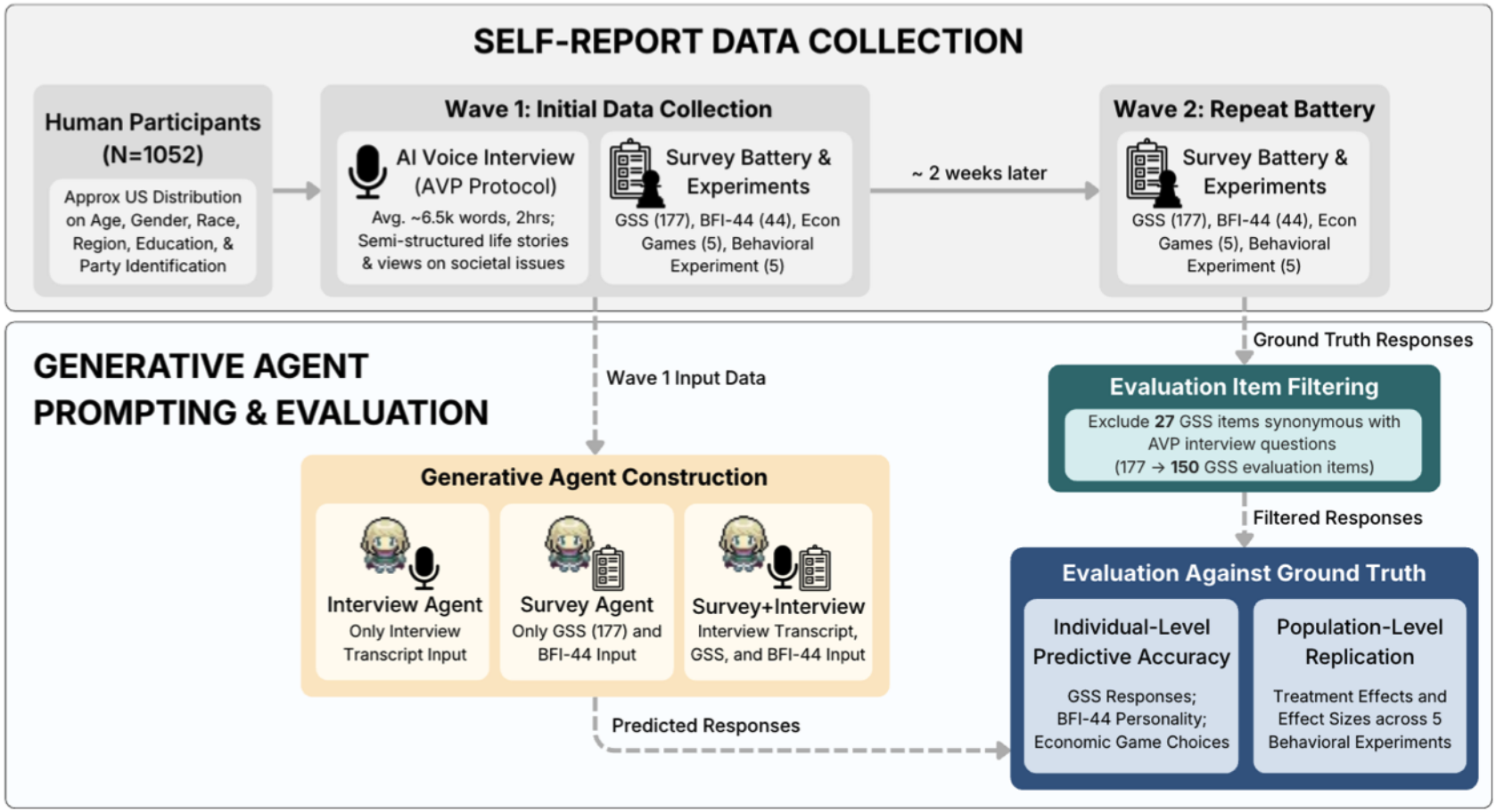
Machine bias



Boelaert, Julien, Samuel Coavoux, Étienne Ollion, Ivaylo Petev, and Patrick Präg. 2025. "Machine Bias. How Do Generative Language Models Answer Opinion Polls?" *Sociological Methods & Research* 54(3): 1156–96.

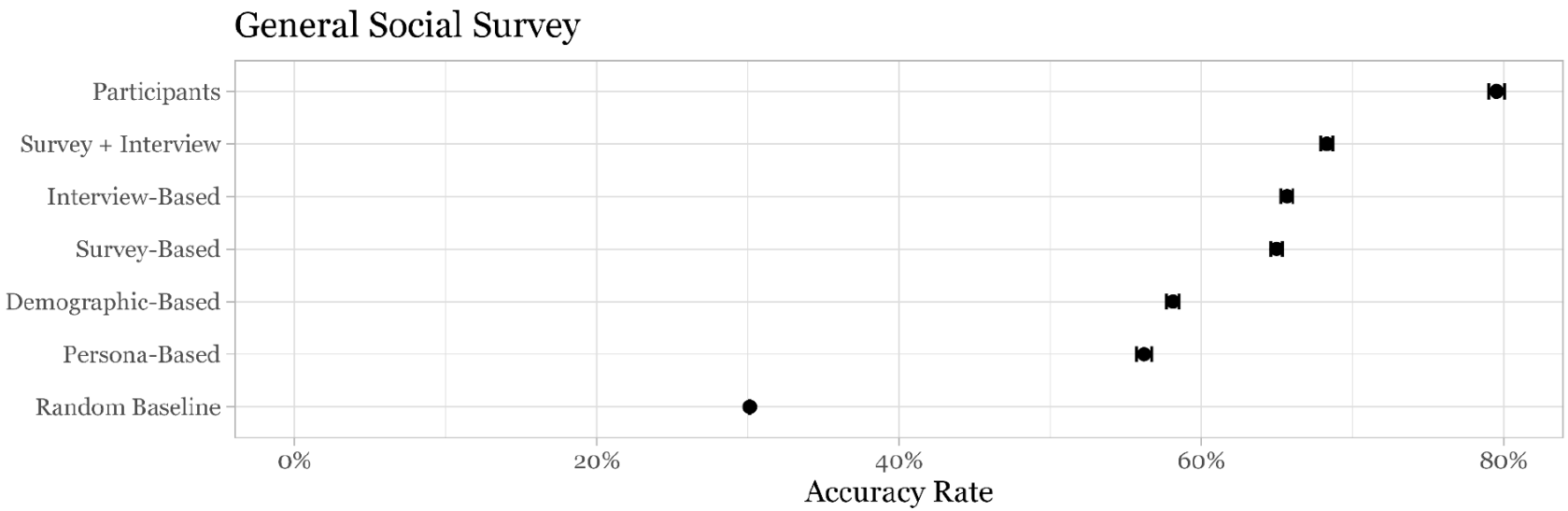
Silicon sampling

Agents from interviews



Silicon sampling

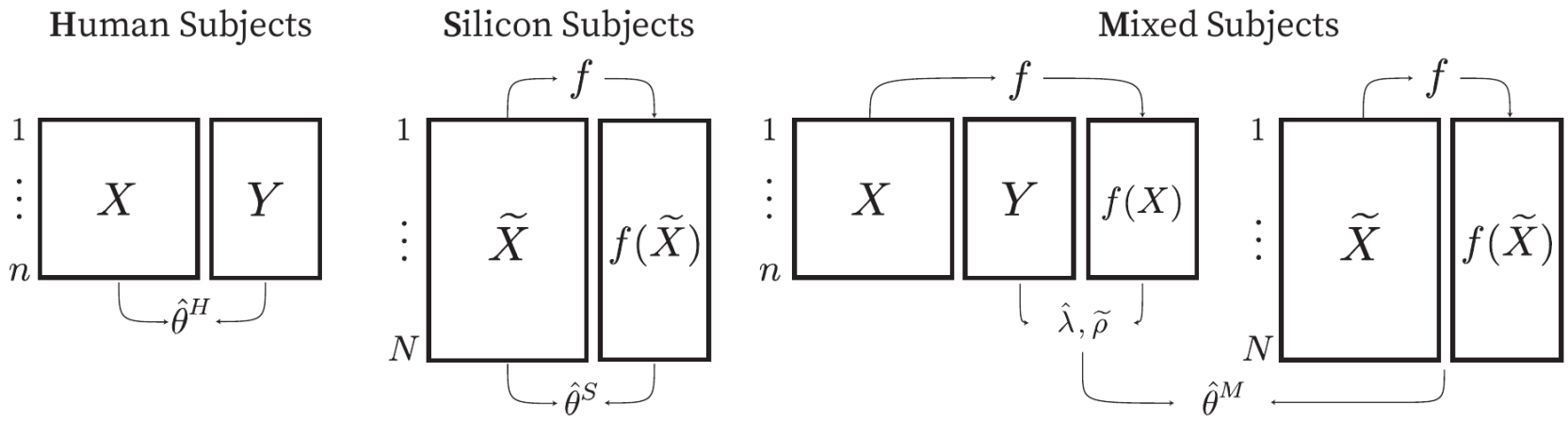
Agents from interviews



Park, Joon Sung, et al. 2026. "LLM Agents Grounded in Self-Reports Enable General-Purpose Simulation of Individuals." *arXiv*:2411.10109.

Silicon sampling

Mixed subjects design



Estimate $\hat{\theta}^H$ with classic inference, e.g. by using OLS to regress observed outcome Y on X.

- ✓ valid estimate
- ✗ precise estimate
- ✗ inexpensive data

Estimate $\hat{\theta}^S$ with classic inference, e.g. by using OLS to regress LLM predictions $f(\widetilde{X})$ on $\widetilde{X}$.

- ✗ valid estimate
- ✓ precise estimate
- ✓ inexpensive data

Estimate $\hat{\theta}^M$ by combining data from human subjects and LLM predictions. Prediction-powered inference (PPI) assesses their interchangeability $\tilde{\rho}$, avoids bias in parameter estimates, and enhances precision through power-tuning $\hat{\lambda}$.

- ✓ valid estimate
- ✓ precise estimate
- ✓ inexpensive data

Broska, David, Michael Howes, and Austin van Loon. 2025. "The Mixed Subjects Design: Treating Large Language Models as Potentially Informative Observations." *Sociological Methods & Research* 54(3): 1074–1109.

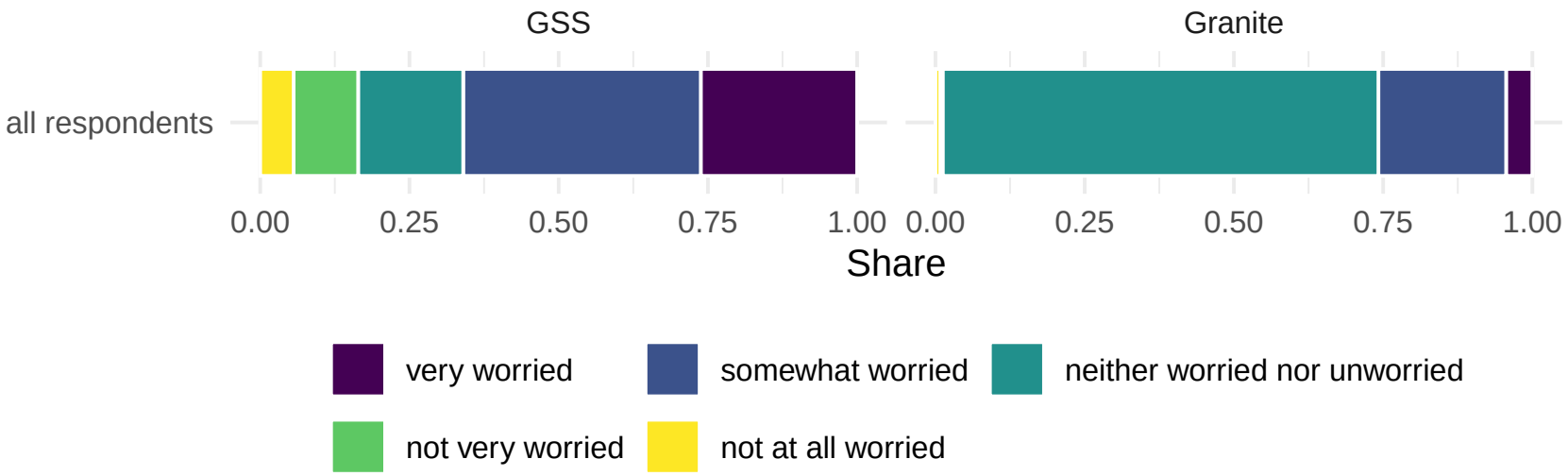
Silicon sampling

Simulating the GSS

- Simulate the answers of 500 GSS 2024 respondents to a question about AI
  - IBM Granite 4.0 micro via `localLLM`
- *System prompt*: a backstory with the respondent's age, sex, race, education, and political views
- *User prompt*: "Overall, how worried, if at all, are you that in the next 10 years machines, computer programs, and Artificial Intelligence (AI) will take over many of the jobs done by humans?" followed by the five options, numbered

Silicon sampling

Answer distribution



Simulated answers have lower variation (SD 0.59 vs. 1.13) and a different distribution

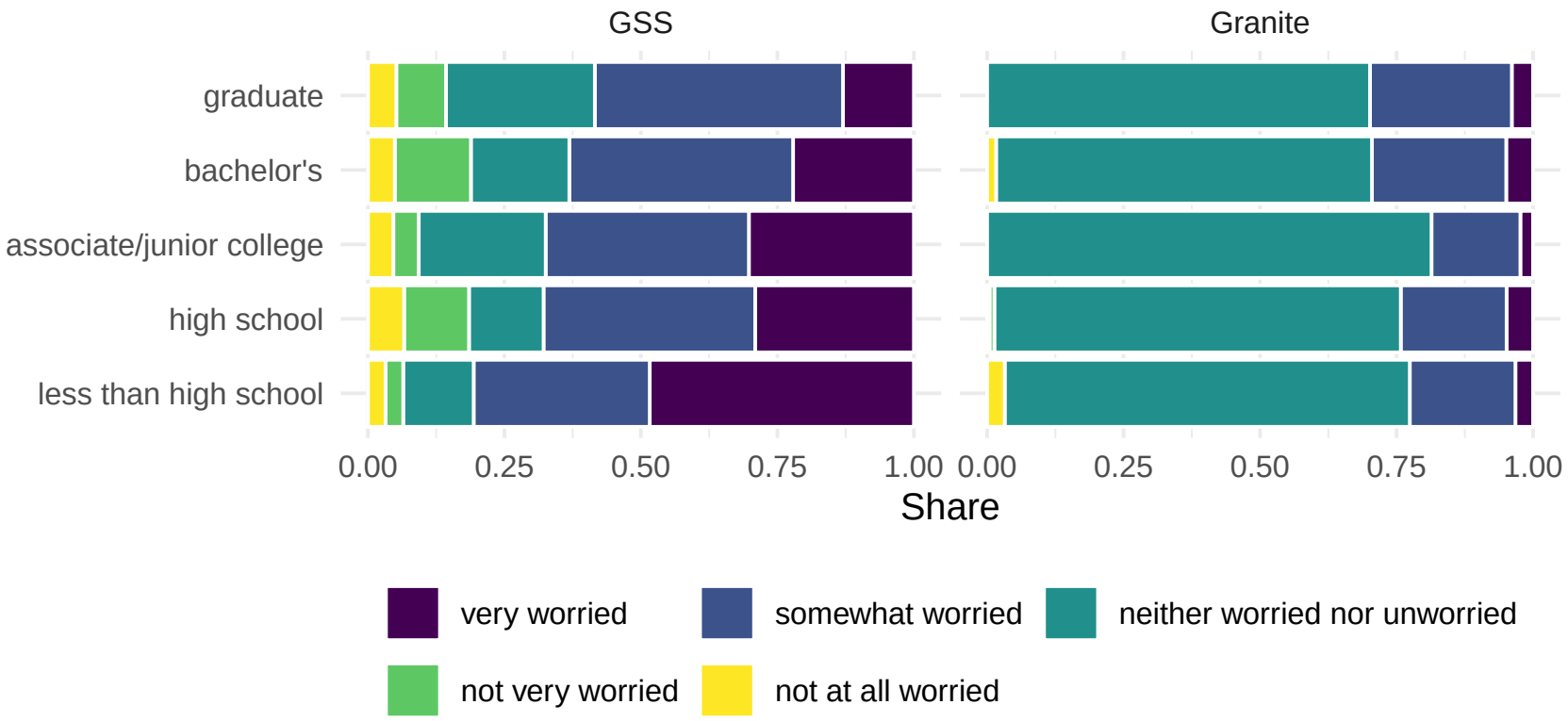
Silicon sampling

By political views



Silicon sampling

By education



Silicon sampling

Exercise

- Edit `code/silicon_sampling.R`
- Choose another GSS question asked in 2024 and change `item` and `question` at the top of the script
- Does the model reproduce the differences between groups? Are the simulated answers less varied than the real ones, or biased?

Agent-based models

Generative agents

- Conventional agents in ABMs follow simple rules
  - Recall the "low-dimensional" realism of Schelling's segregation model
- Generative agents enable more complex simulations, using an LLM to store a record of each agent's experiences, synthesize memories into reflections, and retrieve them to plan behavior

Bail 2024; Park et al. 2023

Agent-based models

Generative agents

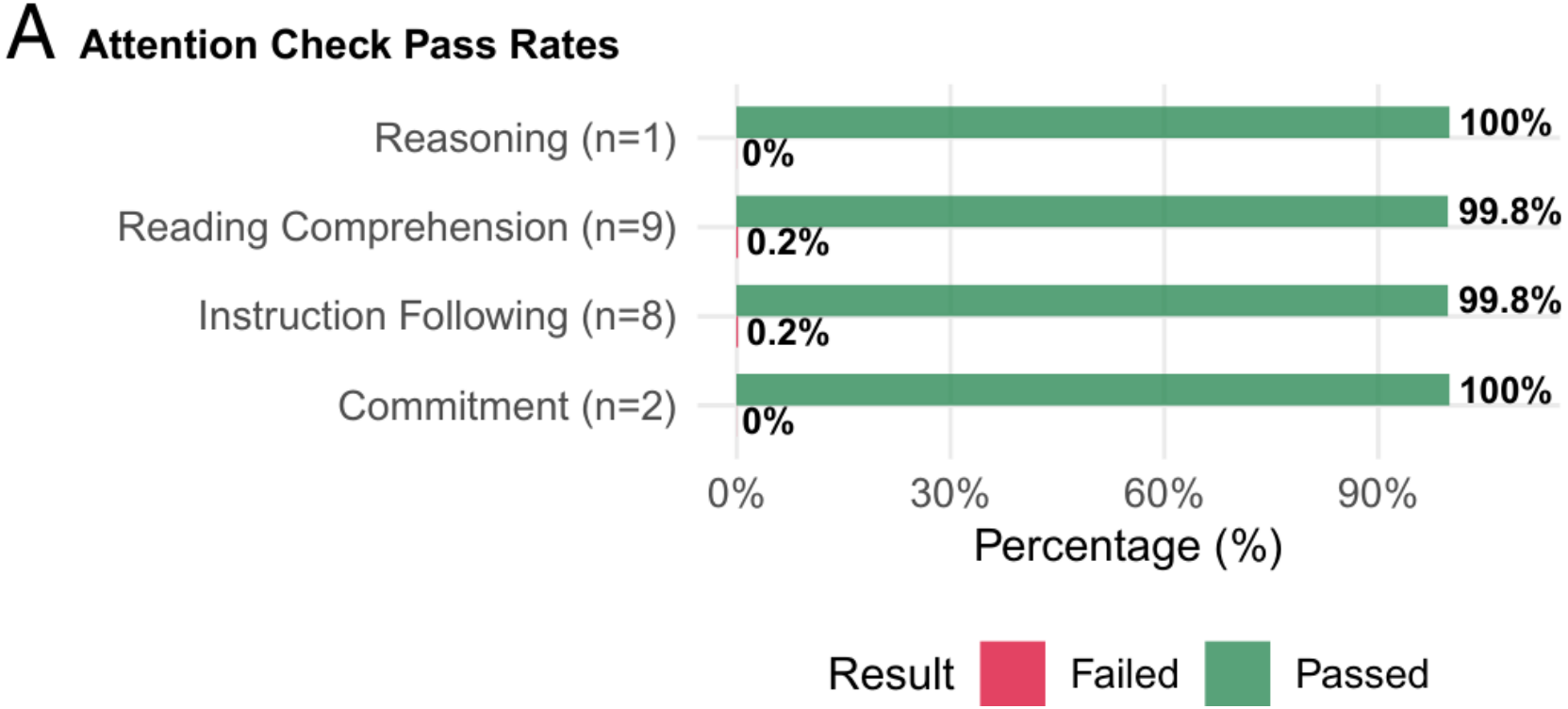


Figure 1: Generative agents are believable simulacra of human behavior for interactive applications. In this work, we demonstrate generative agents by populating a sandbox environment, reminiscent of The Sims, with twenty-five agents. Users can observe and intervene as agents plan their days, share news, form relationships, and coordinate group activities.

Park et al. 2023

AI and online surveys

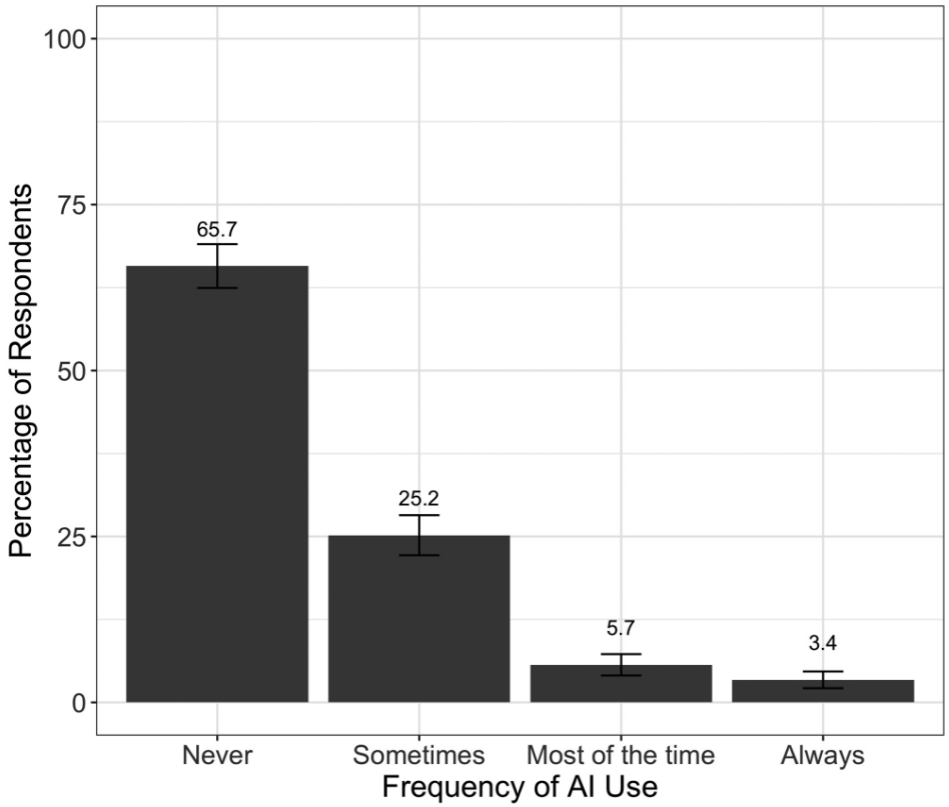
Unwanted synthetic respondents



Westwood, Sean J. 2025. "The Potential Existential Threat of Large Language Models to Online Survey Research." *Proceedings of the National Academy of Sciences* 122(47): e2518075122.

AI and online surveys

Participant use of AI



Zhang, Simone, Janet Xu, and AJ Alvero. 2025. "Generative AI Meets Open-Ended Survey Responses: Research Participant Use of AI and Homogenization." *Sociological Methods & Research* 54(3): 1197–1242.

Human–AI experiments

Human–AI experiments

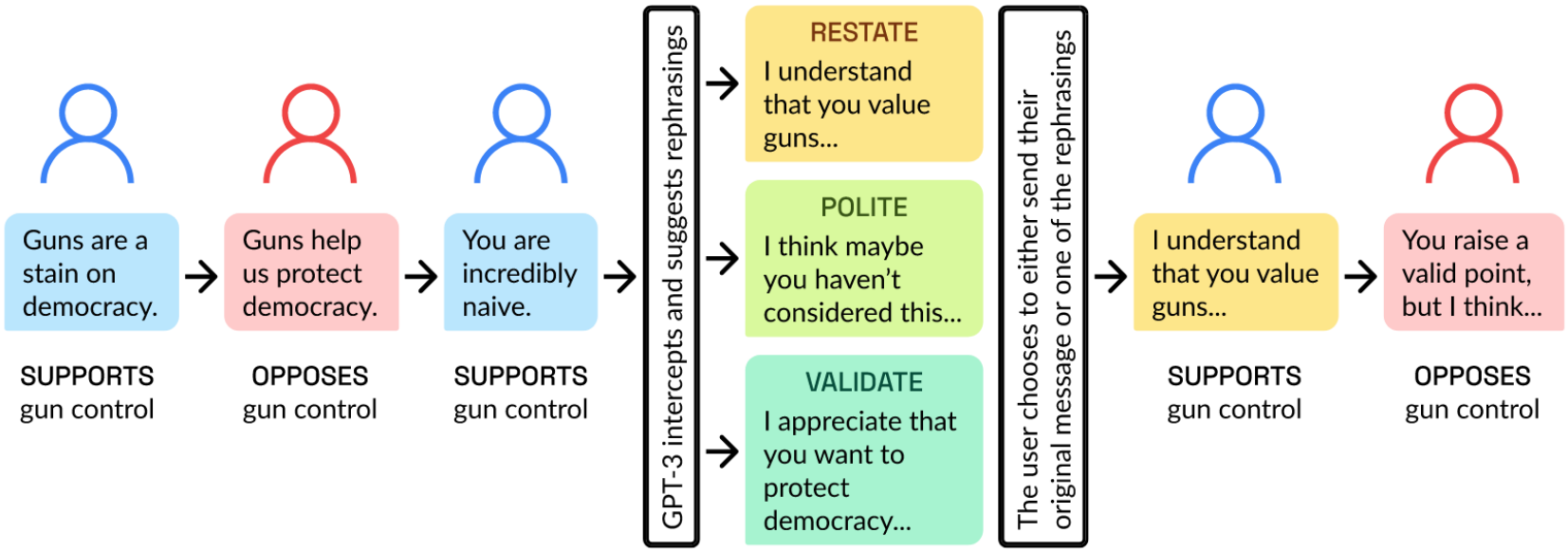
Chat as treatment

- An LLM made real-time, evidence-based recommendations to help participants feel understood in conversations about gun control
- The interventions improved reported conversation quality, promoted democratic reciprocity, and improved tone, without changing policy attitudes

Argyle et al. 2023, *PNAS*

Human–AI experiments

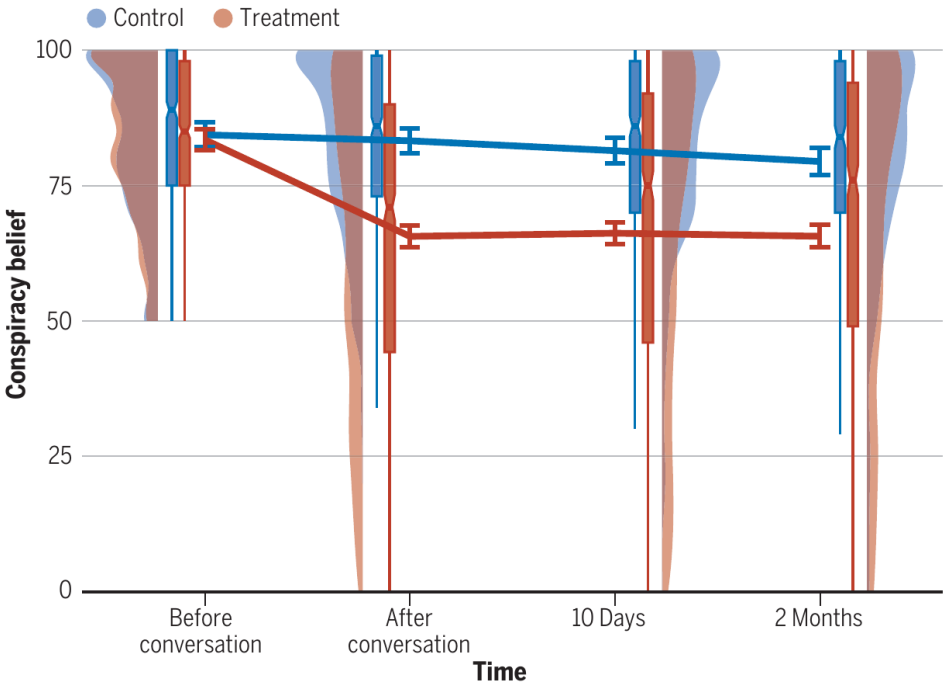
Chat interventions

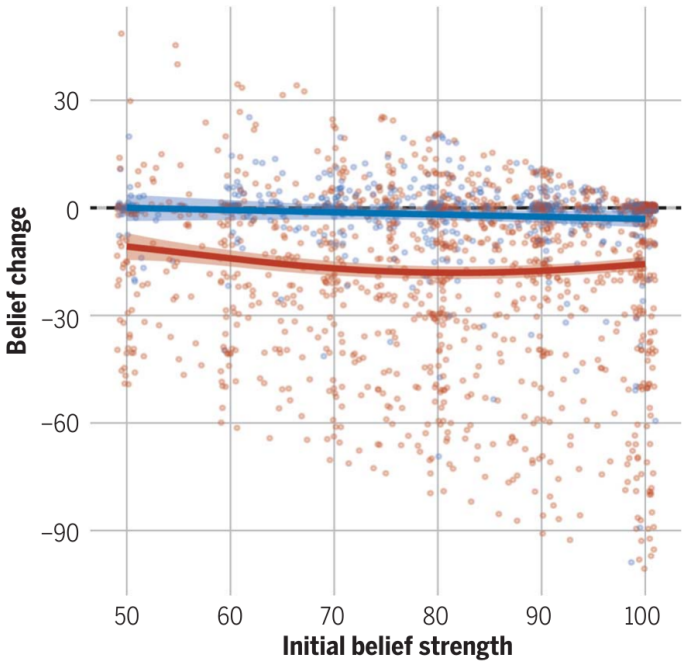


Argyle et al. 2023, *PNAS*

Human–AI experiments

Reducing conspiracy beliefs

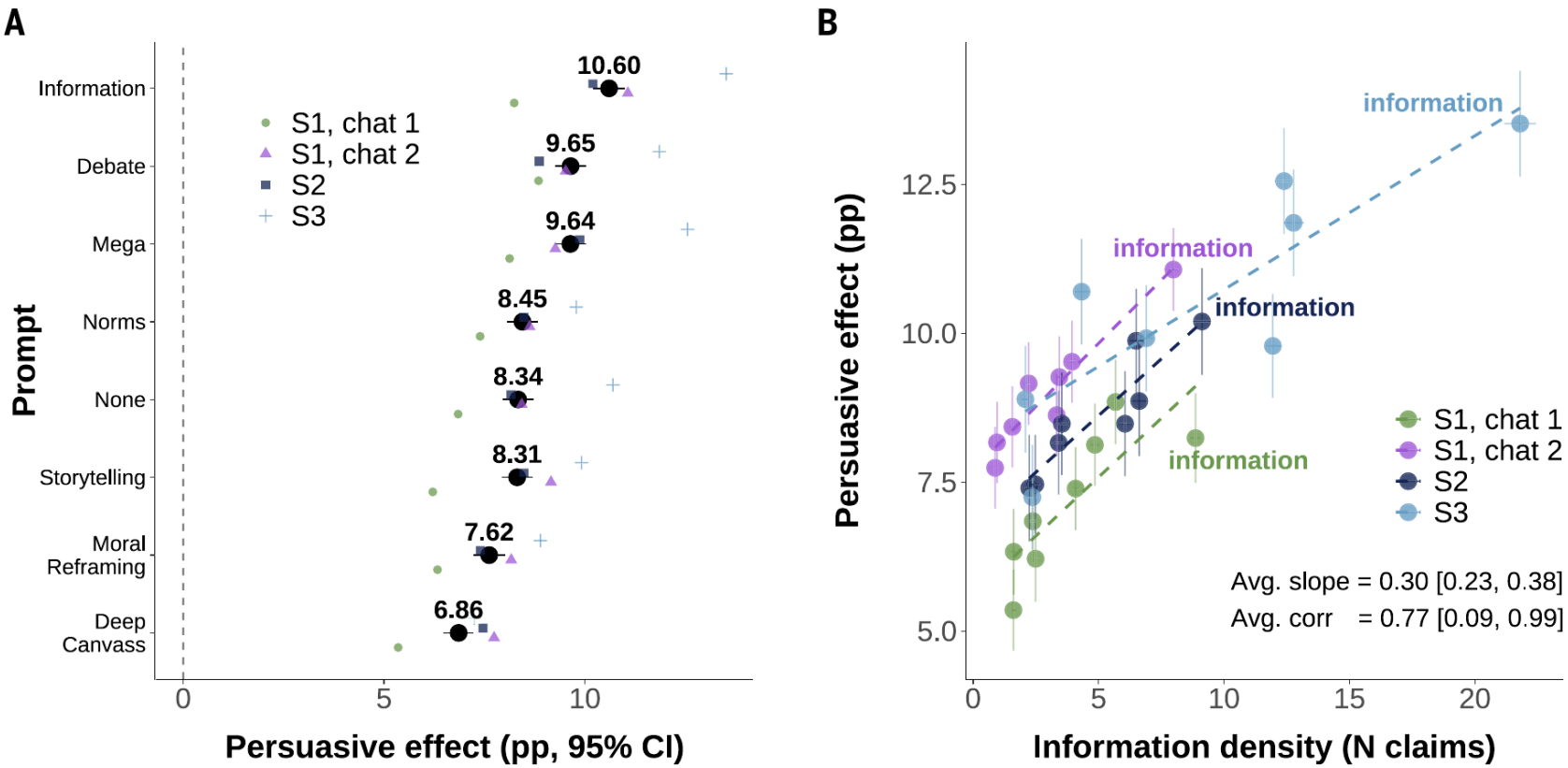




Costello, Thomas H., Gordon Pennycook, and David G. Rand. 2024. "Durably Reducing Conspiracy Beliefs through Dialogues with AI." *Science* 385(6714): eadq1814.

Human–AI experiments

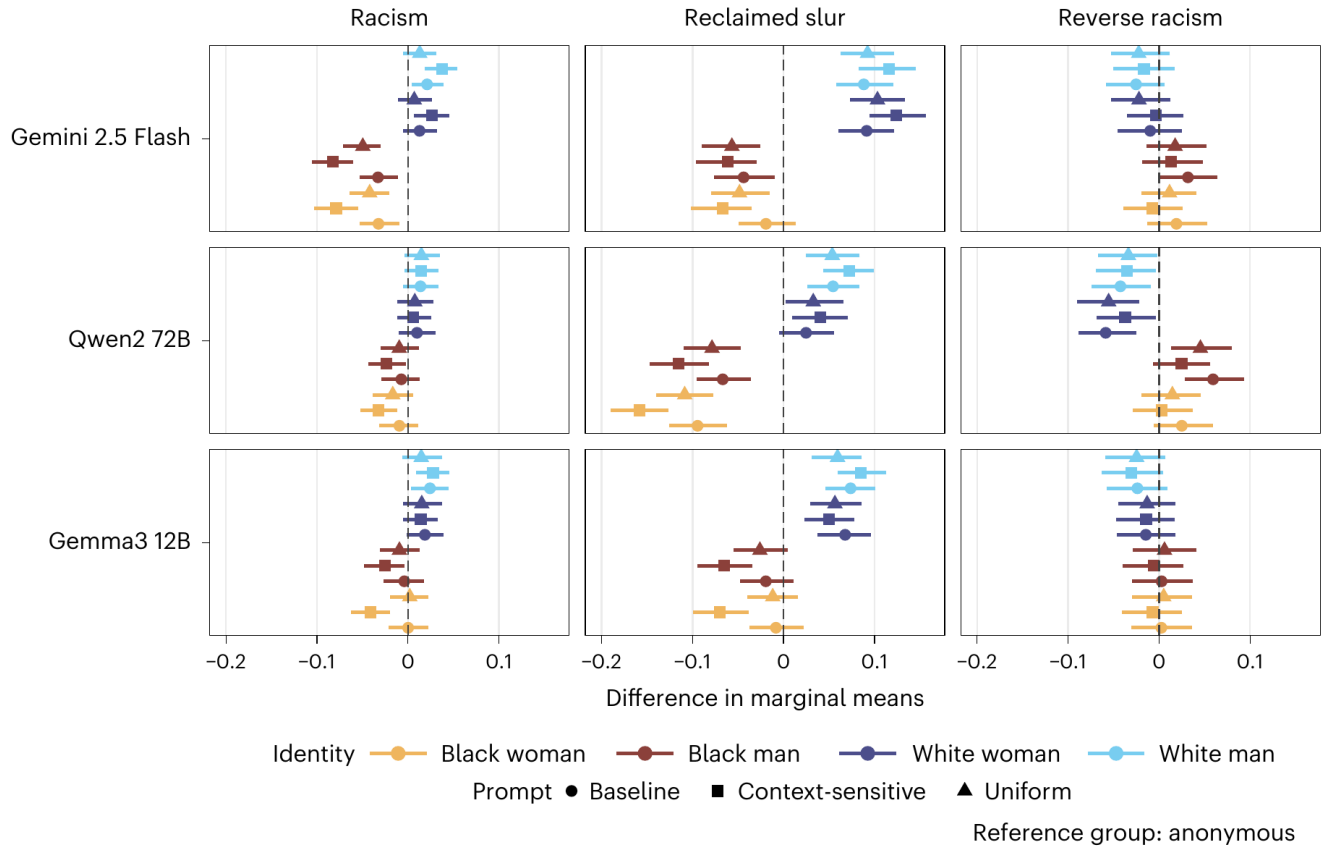
Levers of persuasion



Hackenburg et al. 2025

Human–AI experiments

Auditing AI



Davidson 2026

Human–AI experiments

Ethics

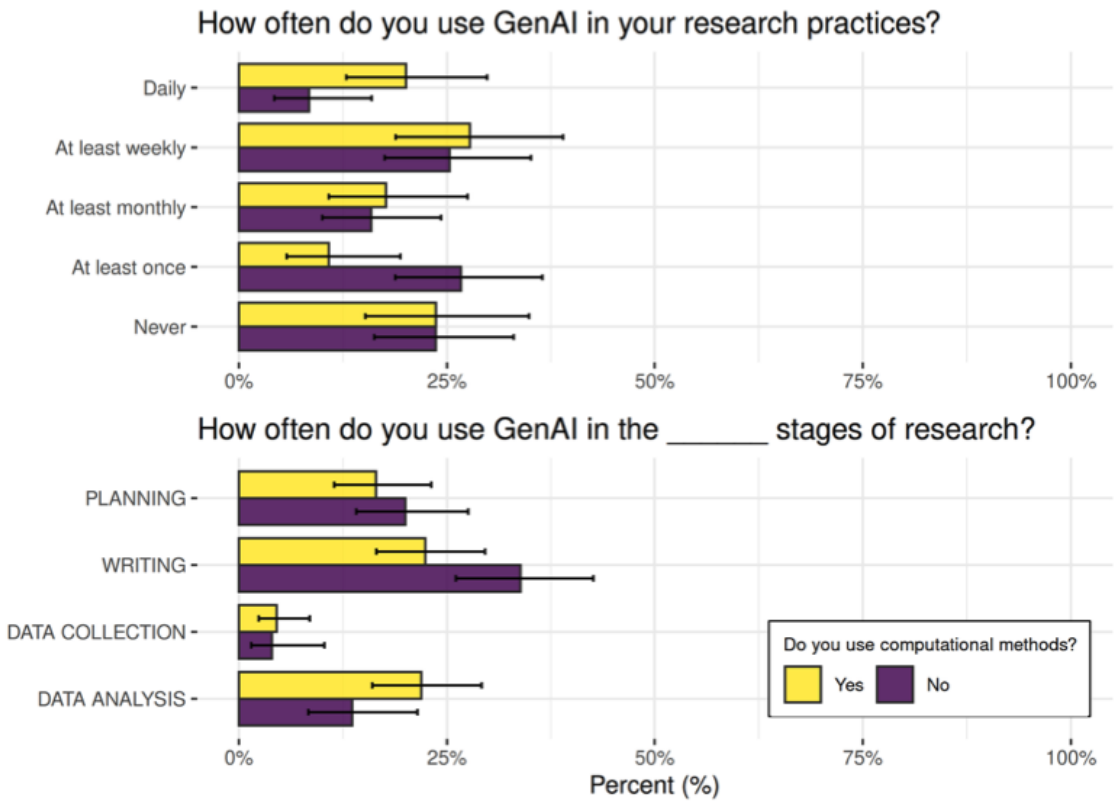
- Should participants always be told they are interacting with AI?
  - Disclosure is critical where participants could be exposed to misinformation or abuse
  - But disclosure makes it harder to tell whether behavior reflects the interaction or attitudes toward AI
- Using AI as a mediator between human participants avoids deception

Bail 2024

Research practice

Research practice

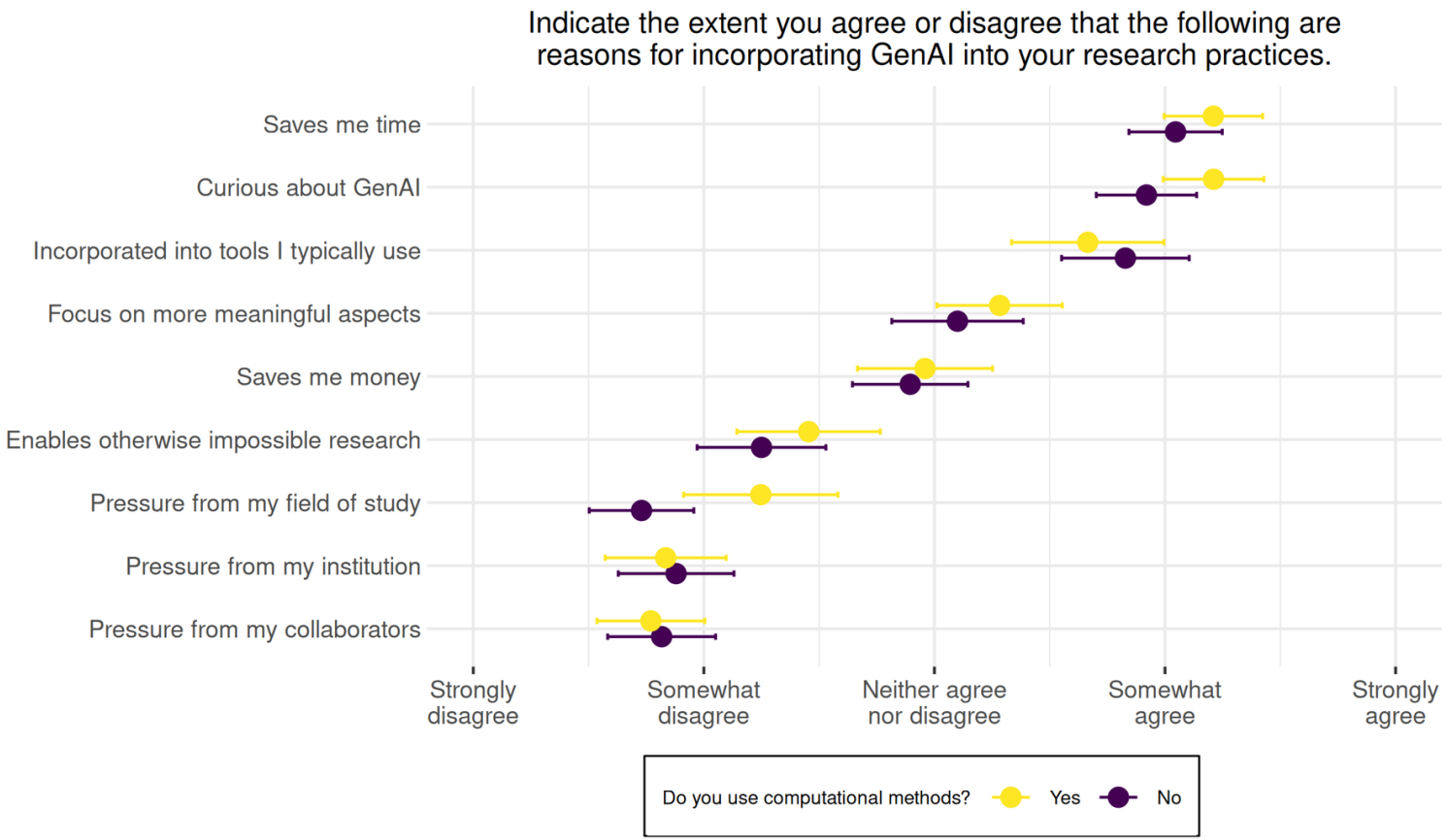
Adoption



Alvero, AJ, Dustin S. Stoltz, Oscar Stuhler, and Marshall A. Taylor. 2026. "Generative AI in Sociological Research: State of the Discipline." *Sociological Science* 13: 45–62.

Research practice

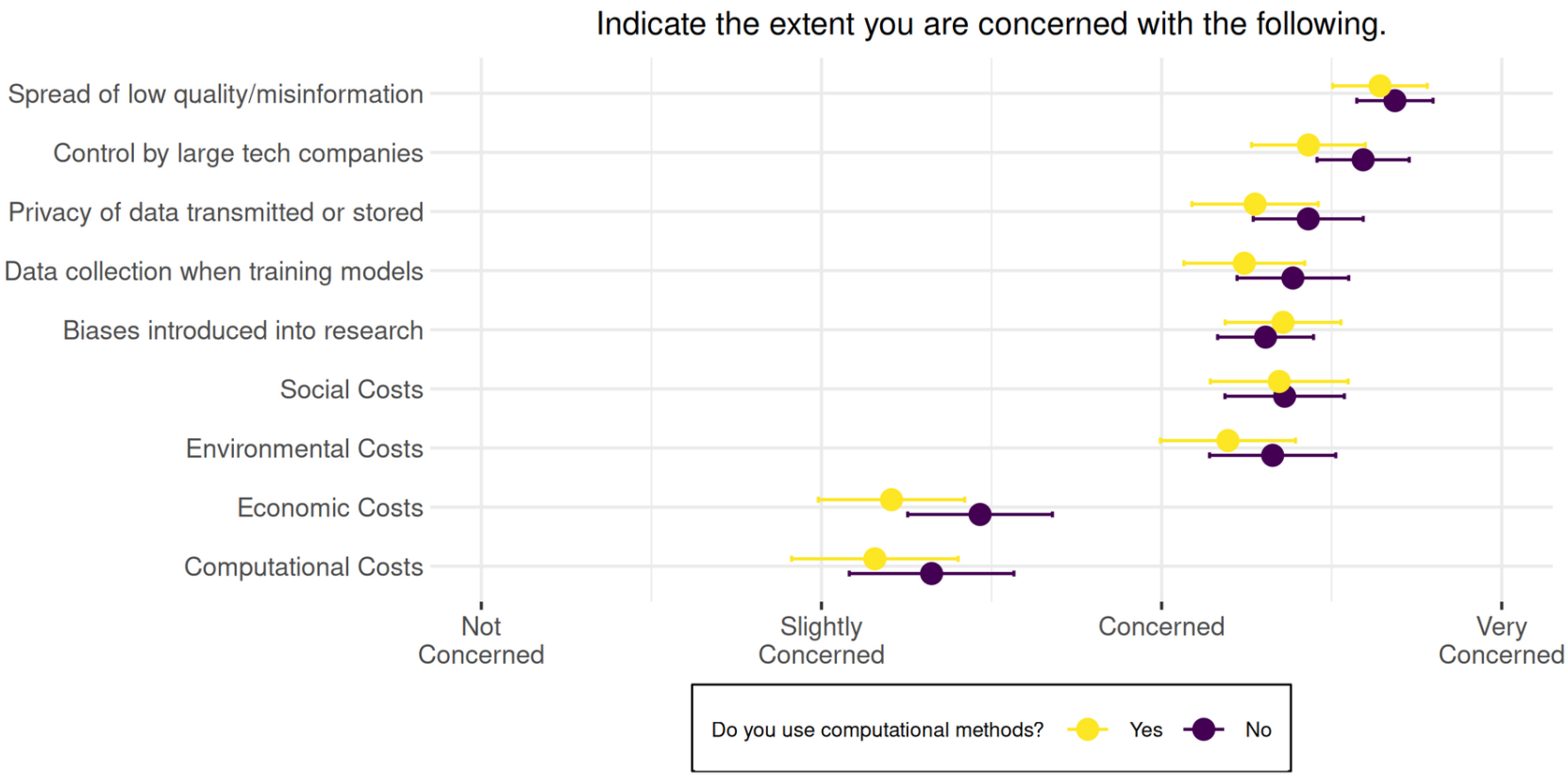
Reasons



Alvero et al. 2026

Research practice

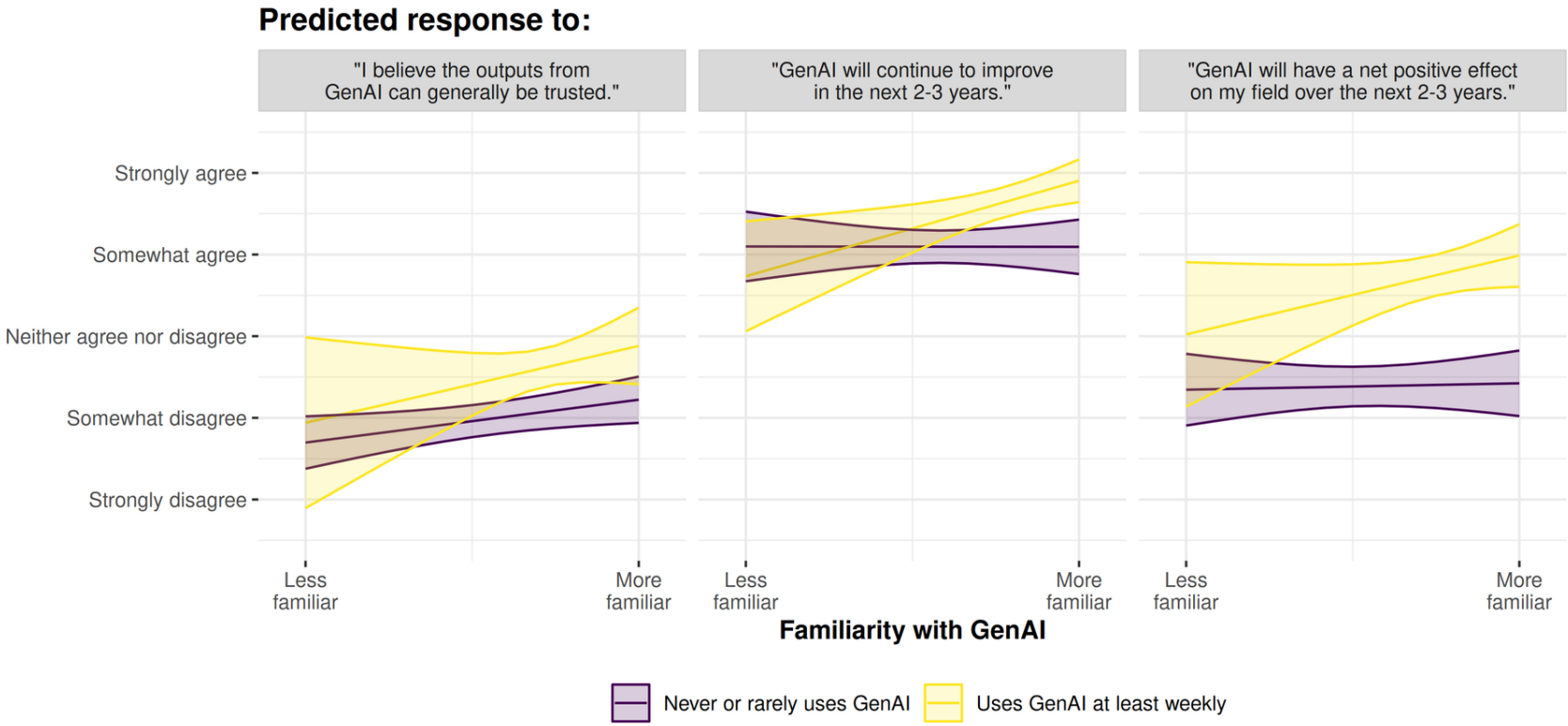
Concerns



Alvero et al. 2026

Research practice

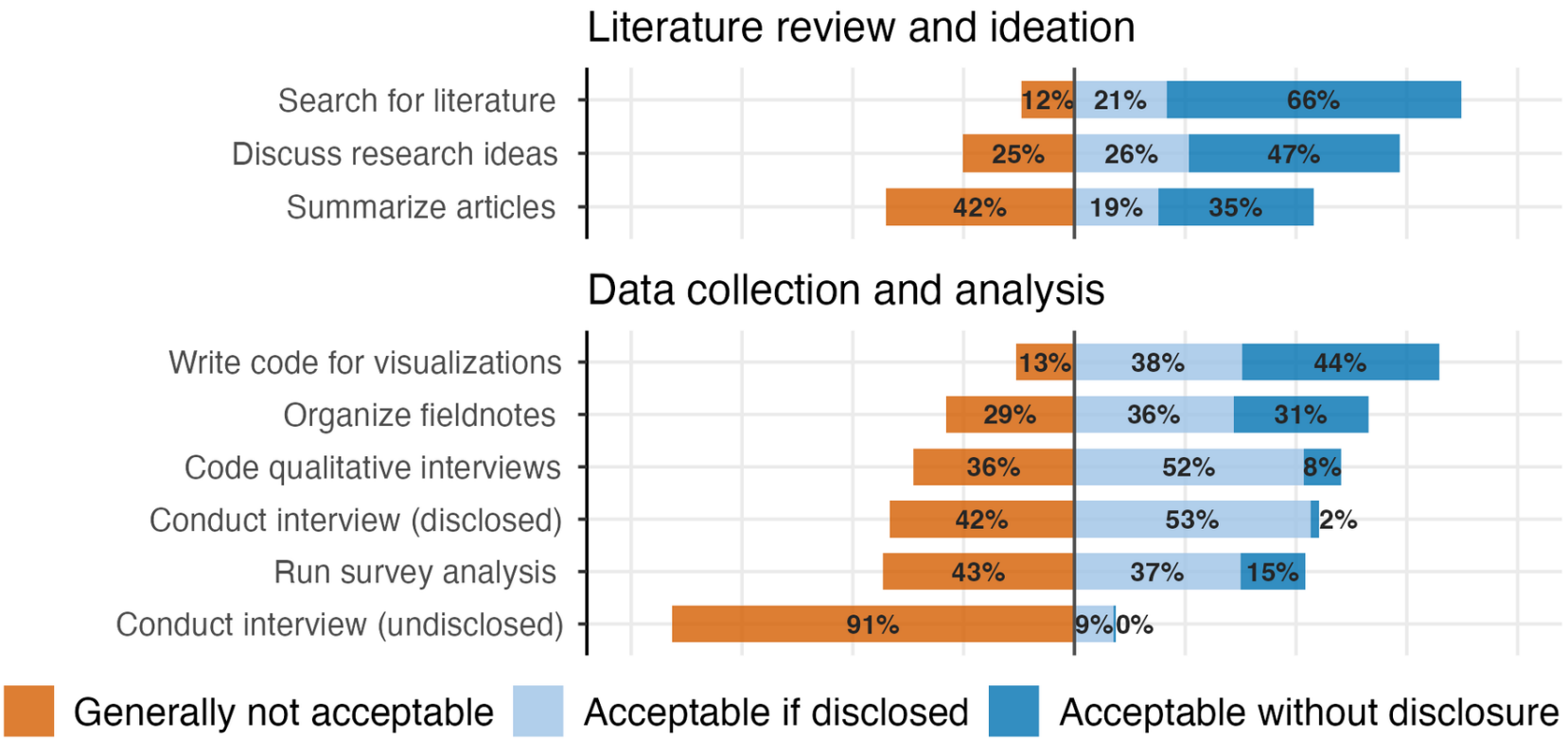
Trust



Alvero et al. 2026

Research practice

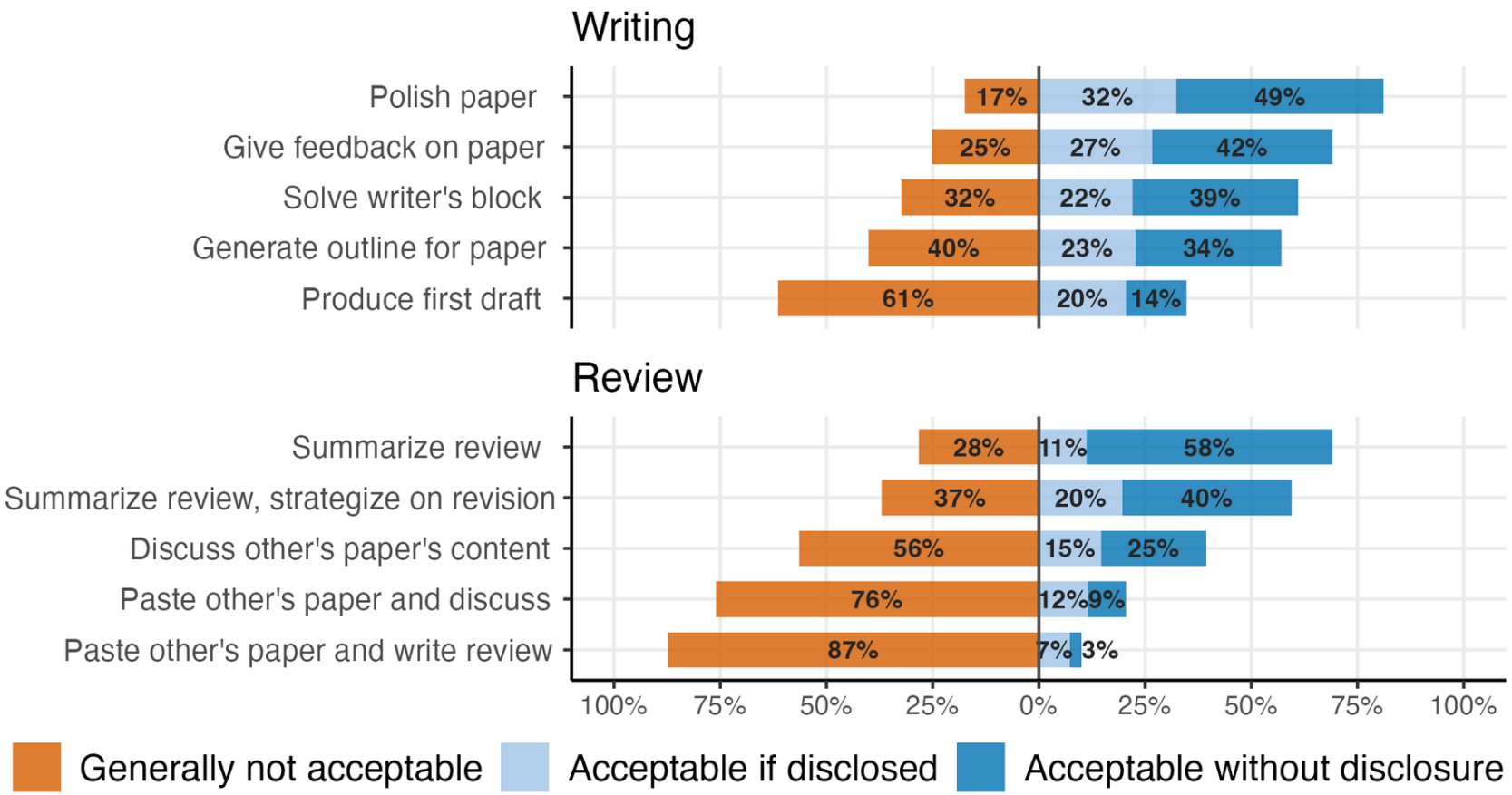
Unstable and evolving norms



Stuhler, Oscar, Dustin S. Stoltz, AJ Alvero, and Marshall A. Taylor. Forthcoming. "Is It Okay to Use Generative AI in Sociological Research?" *Sociological Science*.

Research practice

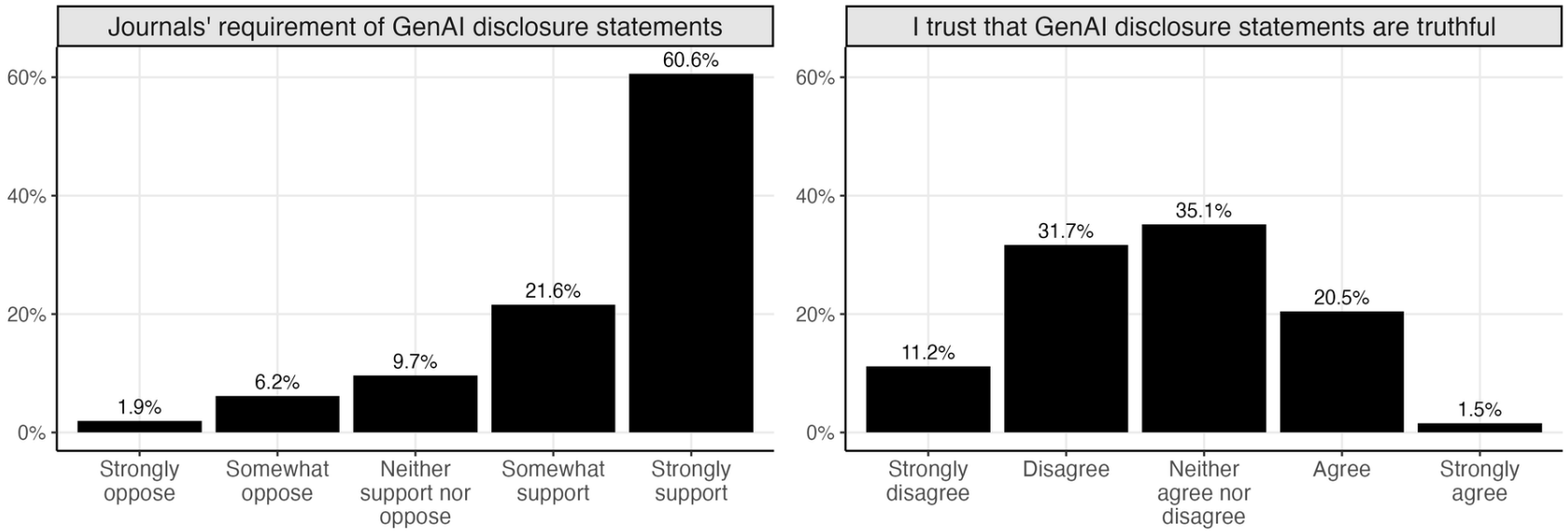
Unstable and evolving norms



Stuhler et al. forthcoming

Research practice

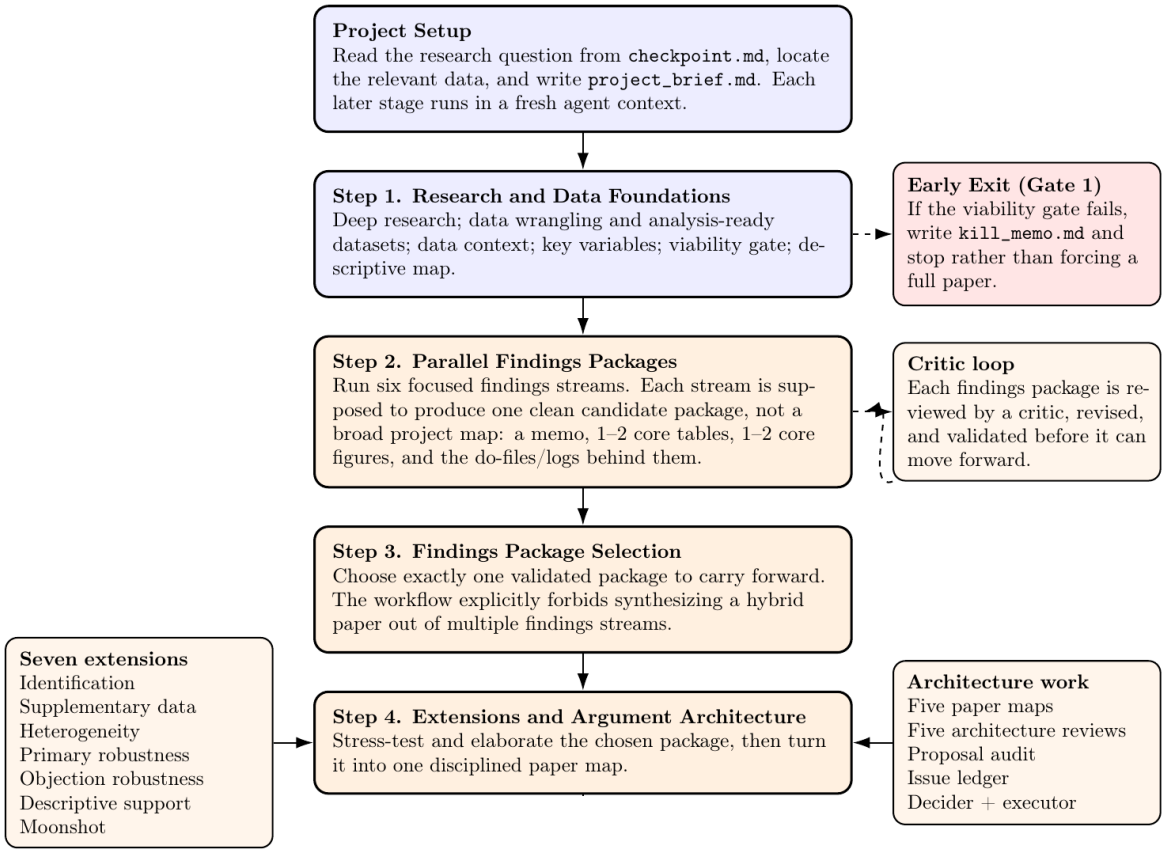
Disclosure requirements



Stuhler et al. forthcoming

Research practice

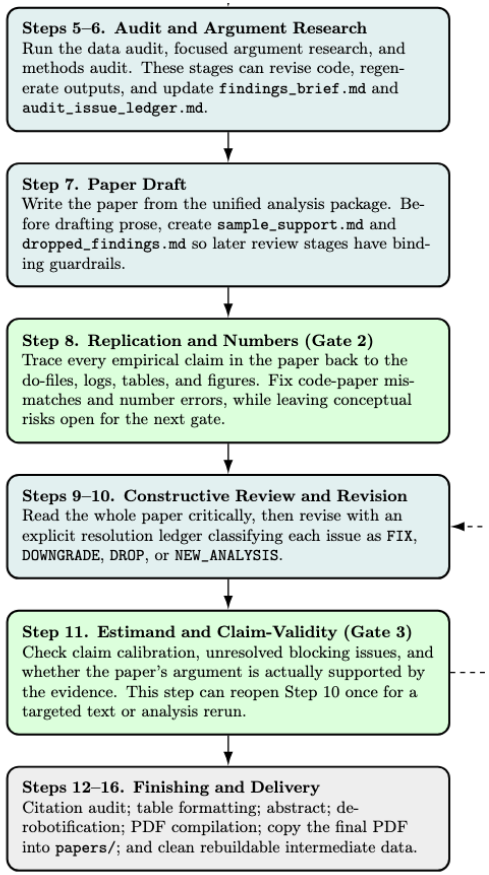
Sociology without sociologists?



Engzell, Per, and Nathan Wilmers. 2026. "The Paper Factory." *SocArXiv*.

Research practice

Sociology without sociologists?



Engzell and Wilmers 2026

Research practice

Sociology without sociologists?

> *So how good are the papers? Quality varies, but many contain discoveries that, with little extra tweaking, would fit well in mid-tier social science journals. As editors, there are papers that we would send out for peer review. There are none that are publishable as-is. Overall, each of these papers has flaws, but they consistently taught us something that we did not already know.*

Engzell and Wilmers 2026. Example papers and code

Summary

- Generative AI can be used for measurement, prompting, and simulation
- LLMs make computational text analysis more flexible and accessible, including for qualitative research
- Silicon sampling and generative agents have potential, but considerable downsides
- LLMs are used as interventions in experiments with human participants and can be audited to understand model behavior
- Debate among sociologists about appropriate uses

Next class

- Computer vision and multimodal models